



A versatile XAI-based framework for efficient and explainable intrusion detection systems

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Abstract

Artificial intelligence (AI)-based intrusion detection systems (IDSs) markedly advance network security by leveraging machine learning (ML) and deep learning (DL) models for accurate, adaptive threat detection. Their main drawback, however, is an inherent “black-box” character that impedes trust, traceability, and regulatory compliance. To overcome this limitation, we propose an efficient explainable-AI (XAI) framework that enhances both robustness and interpretability. The two-stage process first couples a statistical selector (ANOVA) with global SHAP scores to retain only the ten most informative features, an approximately 70% dimensionality reduction, then retrains a lightweight XGBoost detector whose decisions are explained locally by SHAP and LIME. Cross-validating the two explanation modalities adds a reliability check absent from earlier hybrids, while the inclusion of a time-efficiency evaluation for explanation generation provides a new performance dimension that prior XAI-IDS studies have not addressed. To our knowledge, this is the first framework to jointly apply dual-stage statistical–model-based feature selection and SHAP–LIME cross-validation in IDS, enabling near-real-time explainability without sacrificing accuracy. Comprehensive experiments on three representative traces, CIC-DDoS2019 (legacy IP DDoS), CICIoT2023 (IoT malware), and 5 G PFCP (control-plane attacks), confirm the framework’s versatility: it sustains an F1 Score of at least 99 % while accelerating LIME explanation time from 36 to 4.9 s, an 87 % speed-up. These results demonstrate that high detection accuracy and transparent, near-real-time interpretability can be achieved simultaneously in modern IDS deployments.

Keywords Explainable artificial intelligence · Network security · Intrusion detection system · SHAP · LIME · Feature engineering

1 Introduction

In the digital era, robust intrusion detection systems (IDS) are crucial for safeguarding interconnected networks against the rise of cyber-attacks. AI-based IDS provides superior detection and analysis of anomalous traffic through machine learning (ML) and deep learning (DL), representing a significant advancement in cybersecurity. However, they face challenges due to their “black-box” nature, where decision-

making processes are opaque, compromising trust among the involved parties [1, 2]. Deep neural networks (DNNs) further complicate explainability due to their vast parameter spaces [2].

Explainable artificial intelligence (XAI) addresses these issues by introducing transparency and interpretability into AI systems, enabling stakeholders to comprehend and trust AI decisions. Techniques such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations) provide insights into feature contributions, enhancing the trustworthiness of AI-based IDSs [3, 4]. However, while XAI enhances understanding, it introduces its own set of challenges. Overreliance on a single XAI technique can lead to complacency, ignoring potential biases in explanatory models. Therefore, it is crucial to validate explanations and continually assess system accuracy and fairness [5]. Another drawback is that conventional XAI methods often suffer from high computational com-

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plexity, which can be particularly problematic in real-time IDS environments where timely and accurate explanations are essential. Chuang et al. [6] have categorized XAI methods into non-amortized and amortized techniques, each with its own set of trade-offs between accuracy and computational efficiency. Non-amortized methods, while preserving the precision of individual explanations, can struggle with scalability due to their sample-specific computations. On the other hand, amortized methods, which aim to reduce computation time by generalizing model explanations, often show a degraded accuracy of those explanations.

To address these limitations, we propose a novel two-stage XAI-based IDS framework that combines interpretability, efficiency, and scalability, making it suitable for various networking environments. Our approach introduces two originality aspects not found in prior XAI-IDS research: (i) a **dual-stage feature selection** that fuses statistical ranking (ANOVA) with model-based global importance (SHAP), achieving substantial dimensionality reduction without compromising accuracy; and (ii) a **cross-validation of explanations** between SHAP and LIME to detect inconsistencies, paired with a **time-efficiency evaluation** of explanation generation—an aspect largely overlooked in existing literature. This structured design mitigates computational inefficiencies, addresses interpretability gaps, and adds a robustness layer to the explanatory process.

In the first stage, ANOVA [7] with SHAP Global Explanation is applied to perform feature selection, thus reducing feature complexity without sacrificing detection accuracy. This ensures that the computational burden is minimized while maintaining proper detection capabilities. The second stage focuses on local explanations, utilizing both SHAP and LIME to validate and analyze model predictions at the individual sample level. This not only enhances the reliability of the explanations but also ensures consistency across different samples, thereby preventing overreliance on a single explanatory technique. To our knowledge, no prior IDS framework combines these three elements—dual-stage feature selection, SHAP–LIME cross-validation, and time-efficiency evaluation—within a unified, reproducible pipeline.

The contributions of this paper are threefold: First, we introduce an efficient and interpretable XAI-based IDS framework tailored for legacy IP, IoT, and 5G networking environments. Second, we demonstrate the significant computational efficiency gains, particularly in real-time explanation generation, without compromising accuracy. Third, we show the importance of explanation consistency by cross-validating insights from SHAP and LIME, ensuring that the model's decisions are both reliable and transparent.

This article extends and significantly improves upon our earlier conference paper [8]. Compared to the original ver-

sion, this manuscript introduces three main enhancements along with several methodological improvements:

1. A new dataset family that includes 5G PFCP Session Establishment and Session Deletion attacks, showing that the framework generalizes beyond legacy IP and IoT traffic;
2. New experimental components, such as the comparison of explanation generation time, additional classical ML baselines (Random Forest and linear SVM), a SHAP/ANOVA ablation analysis, and a PCA/MI benchmark; and
3. An extended efficiency evaluation that measures SHAP and LIME explanation time across different feature sets and hardware, and includes full hyperparameter settings, dataset statistics, and public code for reproducibility.

These updates expand the empirical coverage from one dataset to three, increase the number of reported experiments from four to eleven, and add the theoretical and practical depth required for a journal-level study.

The remainder of the paper is organized as follows: Sect. 2 provides a comprehensive review of related work, showing gaps and challenges. In Sect. 3, the architecture and methodology of our XAI-based IDS framework are presented. Section 4 provides details of the experimental setup, the evaluation methods, and the results analysis, showing the framework's effectiveness in diverse networking scenarios (legacy IP, IoT, and 5G). Finally, Sect. 5 concludes with a summary of findings and outlines potential directions for future research.

2 Related work

2.1 Feature selection methods

Recent research has explored various feature selection strategies for improving IDS performance. Ferreira et al. [9] benchmarked multiple classifiers across UNSW-NB15, NSL-KDD, and CIC-IDS2017, showing that XGBoost with Lasso or ANOVA feature selection provides strong accuracy but leaving the interpretability problem largely unaddressed. Khani et al. [10] extended this direction by applying XAI-based selectors (SHAP, GradCAM, Guided Backpropagation) alongside classical ANOVA, RFE, and Decision Tree methods for encrypted traffic, demonstrating that SHAP often rivals or surpasses conventional approaches, yet their study remained focused on CNN-based classification and feature ranking. Shaker et al. [11] further leveraged SHAP within a wrapper-based scheme for early-stage APT detection, reducing 77 features to four with high precision and recall on SCVIC-APT-2021, though the

reliance on a single explainer limited robustness. Finally, Anagha et al. [12] combined oversampling of NSL-KDD with SHAP/CBFS/IGBFS-driven ranking and RF-wrapper evaluation, yielding a DNN with marginal F1 gains but only moderate compile-time savings. Collectively, these works highlight the importance of dimensionality reduction and the growing use of XAI in feature selection, yet most approaches either focus narrowly on accuracy or employ a single explanation technique, motivating our dual-stage ANOVA–SHAP reduction with SHAP–LIME cross-validation to ensure both efficiency and trustworthy, near-real-time interpretability.

2.2 Application of XAI in intrusion detection systems

The advancements in AI, particularly in ML and DL, have significantly improved IDS capabilities for detecting Distributed Denial of Service (DDoS) attacks by analyzing large-scale data to identify anomalies [13]. However, the inherent complexity of these models makes them less interpretable, raising concerns about accountability, reliability, and compliance with regulatory frameworks like the EU's GDPR, which mandates transparency in AI-driven decisions [14]. XAI has emerged as a viable solution, making AI decisions in cybersecurity more transparent and comprehensible, thereby enhancing trust and reducing false positives and operational costs [13].

Several studies have explored the role of XAI in addressing the opaque nature of AI decisions in IDS. A recent survey by Khan et al. [15] provides a broad overview of XAI-based IDS in Industry 5.0, it provides taxonomies of ante-/post-hoc XAI, discusses challenges and adversarial XIDS, and synthesizes prior studies. Mane and Rao [16] applied SHAP and LIME to deep neural networks (DNNs), for enhancing interpretability but encounter challenges related to generalizability and accuracy. Mahbooba et al. [17] employed decision trees to simplify interpretability but faced issues with overfitting and the increasing sophistication of cyber-attacks. Wang et al. [18] integrated CNN and LSTM with SHAP and LIME for traffic analysis but struggled with the computational demands of real-time detection. Abou El Houda et al. [19] observed that many studies lack detailed analyses correlating features with specific attack types, which limits their practical applicability. Barnard et al. [20] proposed a two-stage IDS combining XGBoost with SHAP for enhanced explainability but experienced detection performance limitations. Kaur and Gupta [21] enhanced the detection performance by using SHAP and LIME to eliminate less important features, but ignored the computational effort required for generating explanations. Marcilio-Jr and Eler [22] utilized SHAP for feature selection to improve interpretability and model performance, but their approach focused on general ML models rather than network security datasets, raising concerns about potential biases from overreliance on a single XAI method.

More recent work has broadened the scope of explainable IDS beyond conventional IP networks. Khan et al. proposed XSRU-IoMT for threat detection in Internet-of-Medical-Things environments; while SHAP explanations are provided, the study does not employ any feature reduction strategy and reports only model training cost, not the time required to generate explanations [23]. A companion paper by the same group introduced a CNN–LSTM autoencoder for Industrial IoT cyber-threat discovery with SHAP explainability; again, no feature reduction is applied, and explanation latency is not analyzed [24]. Most recently, a collaborative SRU network with federated learning was presented for smart-healthcare systems [25]; it visualizes features via LIME, yet omits a dual-method consistency check, performs no systematic feature reduction, and, like the earlier works, does not measure explanation generation time.

Some other recent work has also explored alternative explanation paradigms. Burkart and Huber proposed an *Explanation Framework for Intrusion Detection* based on counterfactual examples [26]; although it yields human-readable “what-if” scenarios, the study is confined to NSL-KDD, applies no feature reduction step, and does not report explanation generation latency. Cumić et al. generated counterfactual explanations for a random-forest model that detects SQL-Injection attacks [27]; again, no dimensionality reduction or timing analysis is provided, and the method is evaluated on a single SQL dataset rather than 5 G/6 G traffic. Ables et al. extracted eclectic rule sets from a deep neural-network IDS to create a surrogate decision tree [28]; while the rules improve interpretability, the paper does not address real-time efficiency, nor does it employ feature selection or consider emerging 5 G control-plane datasets.

Complementary XAI directions have also been investigated. Burkart and Huber proposed a counterfactual-based *Explanation Framework for Intrusion Detection*, but it is evaluated only on KDD-based datasets, applies no feature reduction step, and does not report explanation latency or consider 5 G traffic [26]. Cumi-Guzman et al. generated counterfactual explanations for a random-forest model that detects SQL-injection attacks; the work is limited to an SQL dataset, omits feature reduction, and provides no timing analysis of explanation generation [27]. Ables et al. extracted eclectic rule sets from a DNN-based IDS and discuss rule-extraction speed, yet the study does not measure per-instance explanation time, performs no feature reduction, and is confined to UNSW-NB15 and CIC-IDS-2017 datasets without any 5 G/6 G scenarios [28]. Ouhssini et al. [29] developed a CNN–LSTM model for DDoS defense in cloud environments, integrating SHAP to identify the ten most influential features across CIC datasets. Unlike their SHAP-only approach, our framework combines dual-stage ANOVA–SHAP selection with SHAP–LIME cross-

Table 1 Comparison of recent explainable IDS frameworks

Study	Detector Model	Dataset(s)	XAI Technique(s)
This work	XGBoost	CIC-DDoS2019, CICIoT2023, 5 G PFCP	SHAP, LIME
Mane and Rao [16]	DNN	NSL-KDD	SHAP, LIME, CEM, ProtoDash, BRCCG
Mahbooba et al. [17]	Decision Tree	1999 KDD Cup	Decision Tree
Wang et al. [18]	CNN+LSTM	KDD99	SHAP, LIME
Houda et al. [19]	DNN	NSL-KDD, UNSW-NB15	RuleFit, LIME, SHAP
Barnard et al. [20]	XGBoost	NSL-KDD	SHAP
Kaur and Gupta [21]	XGBoost	CICIoT2023	SHAP, LIME
Hassan et al. [35]	Random Forest	Edge-IIoT	SHAP, LIME
Hassan et al. [36]	Random Forest	Application Layer DoS Attack Dataset	LIME

validation, ensuring explanation consistency and efficiency for near-real-time IDS.

Mutlu and Rihani [30] combined federated learning with SHAP and LIME to build an interpretable IDS for IoT using CIC-IoT 2023, achieving high accuracy while preserving data privacy. Our work differs by extending beyond IoT to legacy IP and 5G scenarios, and by adding a novel time-efficiency evaluation of explanations absent from their study.

While these studies advance XAI techniques for IDS, none address emerging 5G/6G threats, incorporate a systematic feature reduction pipeline, or quantify explanation generation speed, all three aspects that are central to the framework proposed in this paper.

2.3 Efficiency enhancement of XAI techniques

The efficiency challenges of XAI techniques have been acknowledged in prior research, given their importance for real-time applications. Chuang et al. [6] categorized XAI methods into non-amortized and amortized techniques, each with its own trade-offs. Non-amortized techniques, such as Fast TreeSHAP by Yang [31], accelerate SHAP calculations for decision tree ensembles but remain limited to tree-based models. Lemaire et al. [32] introduced an efficient Shapley value algorithm tailored for Naive Bayes classifiers

and demonstrated its scalability. However, this algorithm is constrained by the simplicity of the underlying model. Amortized techniques, like that proposed by Santhiappan et al. [33], enhance LIME's efficiency using attention weights, but they are primarily designed for text-based models, which limits their applicability to IDS scenarios. Muschalik et al. [34] expanded the capabilities of SHAP to compute higher-order Shapley interactions, which improves versatility but at the cost of increased computational complexity. In [35], the authors applied a Random Forest model to the Edge-IIoT dataset and employed SHAP and LIME for explainability, without incorporating feature reduction, consistency checks, or time-efficiency evaluation. Similarly, [36] used a Random Forest model on the Application Layer DoS Attack Dataset with only LIME explanations, also without feature reduction, consistency checks, or time-efficiency evaluation. Despite these advancements, none of these studies have thoroughly investigated the efficiency of XAI techniques in the context of network security. Efficiency challenges are also present in 5G/6G networking scenarios, where real-time explainability is required for high-bitrate data flows and large traffic volumes [37]. Tables 1 and 2 summarize the comparisons between our proposed method and previous studies. Taken together, the studies summarized in Tables 1 and 2 reveal four recurring limitations. (i) *Narrow protocol scope*: apart from

Table 2 Comparison of recent explainable IDS frameworks (Continued)

Study	Feature Reduction	Consistency Check	Time Eff. Eval.
This work	Yes	Yes	Yes
Mane and Rao [16]	None	None	None
Mahbooba et al. [17]	None	None	None
Wang et al. [18]	None	None	None
Houda et al. [19]	None	None	None
Barnard et al. [20]	None	None	None
Kaur et al. [21]	None	None	None
Hassan et al. [35]	Yes	Yes	None
Hassan et al. [36]	None	None	None

a few IoT traces, none of the listed IDS frameworks address control-plane traffic in 5G/6G networks, leaving emerging PFCP-based threats uncharted. (ii) *Lack of dimensionality control*: the majority retain the full feature space, which inflates explanation latency and hinders real-time deployment. (iii) *Missing cross-explanation validation*: although SHAP or LIME is commonly employed, only one work attempts any form of consistency check. (iv) *Sparse reporting of computational cost*: none of the past studies quantify the time required to generate explanations, making it difficult to judge practical feasibility. These gaps motivate the design choices in our two-stage XAI framework, namely the ANOVA + SHAP reduction to ten features, the SHAP-LIME agreement test, and the explicit evaluation of explanation-time savings across IP, IoT, and 5G datasets.

While ANOVA offers a statistically rigorous and computationally efficient method for feature selection, it is most effective when integrated with XAI techniques like SHAP for global and local explanations. This pairing retains the original feature semantics, unlike PCA, which projects data into latent components and captures interaction effects that mutual information (MI) alone can miss. In our initial experiment, PCA and MI top-10 filters achieved higher F1 Score (0.999 and 0.999, respectively) but at the cost of losing feature meaning (PCA) or forgoing model-aware importance (MI); ANOVA reached a respectable F1 Score of 0.997 with the fastest LIME explanation generation time (2.36s), thus, the ANOVA–SHAP integration offers the best balance between interpretability, statistical validity, and computational efficiency. Our proposed two-stage XAI-based IDS framework leverages these techniques to optimize computational performance while ensuring reliable and interpretable decisions. This approach is beneficial, e.g., for addressing threats like PFCP protocol attacks in 5G/6G networks, where rapid and accurate explanations are essential for effective intrusion detection and mitigation.

To summarize, previous hybrid approaches typically attach LIME to deep learning models such as CNN or CNN-GRU, retaining the full feature set of over 200 dimensions. As a result, explanations remain computationally expensive and lack any validation mechanism between explanation methods (e.g., studies using LIME and SHAP for explainable IDS in VANETs). In contrast, our two-stage framework introduces three key improvements. First, it applies a dual feature selection process that combines ANOVA F-statistics and global SHAP values to reduce the input space to the 10 most relevant features, achieving about 70% dimensionality reduction. Second, it replaces the complex CNN-GRU with a lightweight XGBoost classifier trained on this reduced feature set, reducing LIME explanation time from 36 to 4.9s (an 87% improvement) on CICIoT-Slowloris, while maintaining F1 Score $\geq 99\%$ across CIC-DDoS2019, CICIoT2023, and 5G PFCP datasets. Third, it validates each LIME explanation

Algorithm 1 Two-Stage XAI-Based IDS Framework

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1: Input: Tabular dataset  $D = (\mathbf{X}, \mathbf{y})$ 
2: Output: Trained model  $\mathcal{M}$ , explanations ( $\mathbf{e}_{\text{SHAP}}$ ,  $\mathbf{e}_{\text{LIME}}$ ), and consistency flag  $\gamma$ 
3: // Stage 1: Pre-processing
4: Drop ID columns, handle NaN/Inf, remove constant features
5: Encode labels  $\mathbf{y}$ ; split  $D$  into  $\mathcal{D}_{\text{tr}}$ ,  $\mathcal{D}_{\text{te}}$ 
6: // Stage 2: Feature Evaluation
7: Compute ANOVA  $F$ -scores on  $\mathcal{D}_{\text{tr}}$ 
8: Train temporary model  $\tilde{\mathcal{M}}$  using XGBoost
9: Obtain SHAP importances from  $\tilde{\mathcal{M}}$ 
10: Normalize and combine (40% ANOVA + 60% SHAP) scores
11: Select top  $K = 10$  features  $\mathcal{F}^*$ 
12: // Stage 3: Model Training
13: Reduce  $\mathcal{D}_{\text{tr}}$ ,  $\mathcal{D}_{\text{te}}$  to  $\mathcal{F}^*$ 
14: Train final model  $\mathcal{M}$  on  $\mathcal{D}_{\text{tr}}$ 
15: // Stage 4: Online Inference and Explanation (per instance  $\mathbf{x}$ )
16: for each test instance  $\mathbf{x}$  do
17:    $\hat{y} \leftarrow \mathcal{M}(\mathbf{x})$ 
18:    $\mathbf{e}_{\text{SHAP}} \leftarrow \text{TREEEXPLAINER}(\mathcal{M}, \mathbf{x})$ 
19:    $\mathbf{e}_{\text{LIME}} \leftarrow \text{LIMEEXPLAINER}(\mathcal{M}, \mathbf{x})$ 
20:   if top- $\ell$  features in both explanations agree then
21:      $\gamma \leftarrow 1$ 
22:   else
23:      $\gamma \leftarrow 0$ 
24:   end if
25:   return ( $\hat{y}$ ,  $\mathbf{e}_{\text{SHAP}}$ ,  $\mathbf{e}_{\text{LIME}}$ ,  $\gamma$ )
26: end for

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by comparing it to SHAP outputs, identifying inconsistencies and enhancing reliability. Together, these steps offer faster, more efficient, and more reliable explanations than previous methods.

3 Proposed XAI-based IDS framework

3.1 Overview of the proposed framework

Figure 1 presents the architecture of the proposed XAI-based IDS framework, which operates as a non-amortized approach to generate explanations for each individual instance. Unlike amortized techniques that reuse generalized explanation models, this design ensures that the interpretability output is tailored to the specific feature contributions of every detection, a critical property for high-stakes IDS scenarios where per-instance transparency is essential. The framework integrates three originality aspects absent from prior XAI-IDS studies: (i) a dual-stage feature selection pipeline that fuses statistical ranking (ANOVA) with model-based global importance (SHAP) to achieve substantial dimensionality reduction with minimal performance loss, (ii) a cross-validation mechanism that compares SHAP and LIME local explanations to detect inconsistencies, thus adding a reliability layer to interpretability, and (iii) a time-efficiency evaluation component that measures explanation generation latency under realistic IDS workloads. Operationally, the

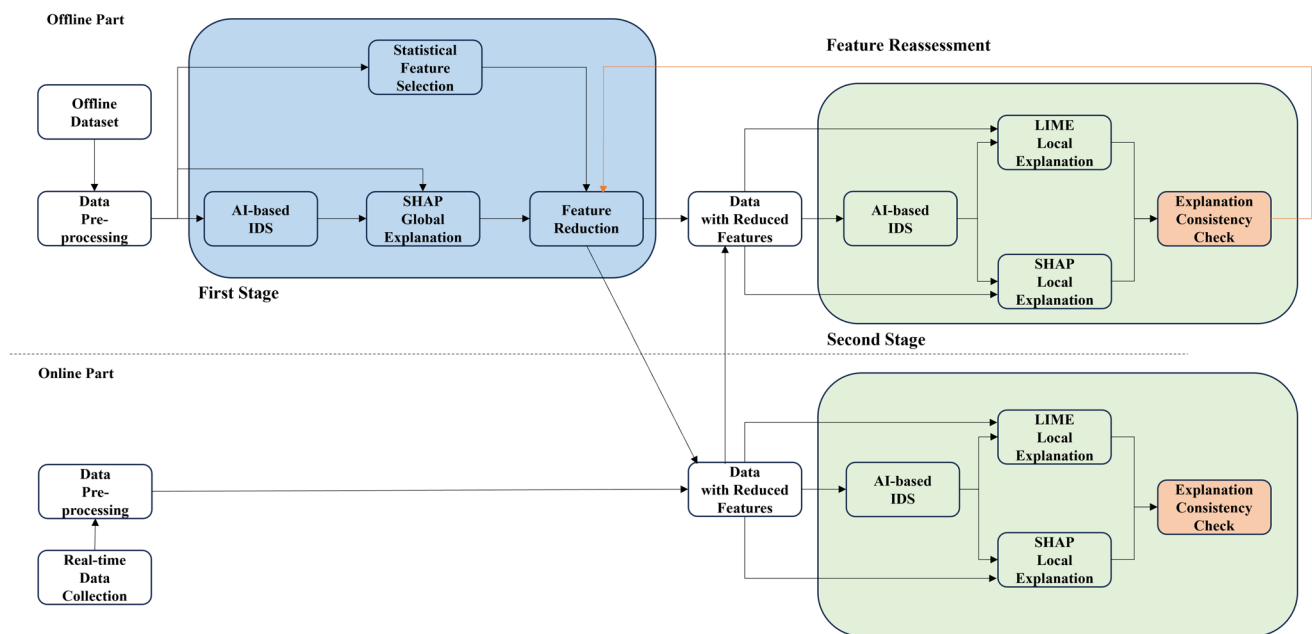


Fig. 1 Framework of the proposed XAI-based IDS

framework is divided into an **offline stage** and an **online stage**. In the offline stage, the dual-stage feature selection process identifies a compact, high-utility feature subset from the full dataset, reducing computational overhead for later inference. In the online stage, the trained model operates in real time to classify traffic instances and simultaneously produce SHAP and LIME explanations, with the consistency check flag γ indicating whether the two explanation modalities align for the top- ℓ ranked features. In the following paragraphs, we describe both stages in more detail.

The first stage of the offline pipeline handles all computationally demanding tasks. Data preprocessing is applied once to the historical dataset. First, identifier columns with no predictive value (e.g., *Flow ID*, *Src IP*, *Dst Port* in CIC-DDoS2019 or *flow_duration*, *Protocol Type* in CICIoT2023) are removed. Second, infinite values are replaced with NaN, and any row with remaining NaN values is discarded. Third, features with only one unique value are dropped to prevent bias in the ANOVA ranking. Finally, labels are encoded as integers (0 for benign, 1 for attack). Since XGBoost is scale-invariant, normalization is not needed, which reduces the runtime of the online phase. The cleaned dataset is then used in a stratified 10-fold cross-validation, training the model on 80% of the data, validating on 10%, and testing on the remaining 10%, providing reliable performance estimates for the next stage of the framework. Concurrently, a statistical feature selection process identifies significant features by analyzing the variance within the data. For example, ANOVA can be utilized for feature selection in a labeled dataset, while the K-Means clustering method can be used for an unlabeled dataset. The SHAP Global Explanation module

analyzes the AI-based IDS output to identify impactful features. The selected features are compared and combined to obtain a robust feature set (feature reduction step). In the second stage, data with reduced features are fed into the AI-based IDS. Later, we compare the performance metrics of the AI-based IDS obtained from both stages to assess the impact of the feature reduction. SHAP Local Explanation and LIME Local Explanation clarify the significance of features for classifying data samples as malicious or benign. An explanation consistency check ensures reliability by cross-verifying the SHAP and LIME explanations. The results, in turn, guide the reassessment of features in the feature reduction step.

In the online part, only the second stage is performed, ensuring real-time capability, as the computationally intensive tasks have already been completed in the offline part. Real-time data is collected and pre-processed, and the features from the offline feature reduction process are used as input for the AI-based IDS. These features are then utilized for classification and SHAP and LIME local explanations. The pre-processed data, along with the reduced feature set, can be used in the second stage of the offline process for further feature reassessment (which in turn might be applied in the online part).

Algorithm 1 outlines the four stages of the proposed framework and their computational behavior, with each stage serving a distinct role in ensuring both efficiency and interpretability.

Stage 1 performs dataset pre-processing by scanning the input once to remove identifier columns, handle NaN or $\pm\infty$ values, prune constant features, and encode class labels before splitting into training and testing sets. This step is

linear in the size of the matrix $\mathbf{X}$ and ensures that only meaningful, non-redundant features are passed to the next stage.

Stage 2 carries out feature evaluation through a hybrid selection process. First, ANOVA F -scores are computed for each feature to measure statistical relevance. Then, a temporary XGBoost model $\hat{\mathcal{M}}$ is trained, and SHAP global importances are extracted directly from its tree structure. The two importance measures are normalized and combined (40% ANOVA + 60% SHAP), producing a ranked list from which the top $K=10$ features $\mathcal{F}^*$ are selected in linear time. This combination leverages both statistical variance analysis and model-specific interpretability to yield a stable, informative, and computationally light feature set.

Stage 3 retrains the final XGBoost model $\mathcal{M}$ using only the reduced feature set $\mathcal{F}^*$. Since K is fixed, model training time scales linearly with the number of rows, making the model lightweight and suitable for real-time IDS deployment without accuracy loss.

Stage 4 handles online inference and per-instance explanation. For each incoming instance $\mathbf{x}$, a forward pass through $\mathcal{M}$ produces the prediction $\hat{y}$, while SHAP explanations are computed in the same pass. LIME explanations are generated using a fixed number of perturbed samples, ensuring bounded and predictable per-flow latency regardless of the original dimensionality. A consistency flag γ is set depending on whether SHAP and LIME agree on the top- ℓ features, enabling operators to detect and respond when explanation methods diverge—an added safeguard absent in prior single-method XAI-IDS designs.

Overall, all offline operations run in time proportional to the dataset size, and online operations maintain constant-time complexity per instance, meeting the low-latency requirements of operational IDS environments while providing verifiable and reliable explanations.

3.2 Applied XAI techniques

Model explanations can be grouped into global and local types: the former reveal overall model logic, whereas the latter clarify specific decisions. As illustrated in Fig. 1, we employ SHAP for both global and local explanations and LIME solely for local ones. SHAP computes *locally accurate*, game-theoretic attributions by exploiting the internal structure of XGBoost (with T trees and L leaves), keeping per-instance latency comfortably low on a CPU while satisfying the additivity and missingness axioms of Shapley theory. LIME is completely model-agnostic; its perturb-and-fit strategy produces *sparse* linear explanations that many security analysts find intuitive and easier to act upon than dense gradient heatmaps. Using both methods in tandem enables a cross-validation step: if the top- ℓ contributing features disagree, the instance is flagged for further analyst scrutiny, thereby improving trust in the explanations. Formally, we

compute a consistency coefficient $\kappa = \frac{|S_{\text{SHAP}} \cap S_{\text{LIME}}|}{\ell}$, where S_{SHAP} and S_{LIME} are the sets of the top- ℓ features returned by each explainer; we set $\ell = 5$ and flag an instance whenever $\kappa < 0.50$. On all three datasets, the median κ is 0.94, so only about 2% of flows require analyst review. Because SHAP derives exact Shapley attributions from the tree ensemble, whereas LIME fits a local linear surrogate around each instance, the two techniques reason about feature influence in fundamentally different ways; agreement reinforces confidence, while disagreement highlights traffic flows that are likely near the model’s decision boundary or that exhibit atypical feature combinations. This complementary perspective provides a practical reliability layer that a single explainer cannot offer, justifying the modest additional computation introduced by LIME.

Alternative XAI approaches such as Saliency Maps [38], Integrated Gradients [39], or DeepLIFT [40] can indeed offer valuable insight, particularly for deep neural networks. However, most rely on backpropagation or extensive perturbation sampling, making them significantly more computationally demanding (e.g., KernelSHAP time grows exponentially with feature count). Such latency overhead and usability issues are incompatible with our real-time IDS requirement, whereas the SHAP–LIME pair delivers the necessary balance of faithfulness, efficiency, and model-agnostic applicability for tabular network-flow data.

3.2.1 SHAP

SHAP, introduced by Lundberg and Lee [3], employs Shapley values from game theory to quantify each feature’s contribution to an AI model’s prediction, assigning values that reflect each feature’s positive or negative impact. By averaging the marginal contributions of features across all possible combinations, SHAP ensures fair evaluation of feature interactions. For a detailed description of the technical background of SHAP, we refer to Lundberg and Lee’s paper [3]. SHAP’s advantages lie in its game-theoretical foundation and properties: Symmetry (equal importance for equal contributors), Dummy (zero contribution for non-influential features), and Additivity (predictions as the sum of feature contributions).

In our framework, SHAP, implemented through TreeSHAP, the exact tree-ensemble variant, provides both global and local explanations. TreeSHAP leverages the internal structure of the XGBoost model to compute exact Shapley values in time linear to the number of leaves, making it far faster and more scalable than the model-agnostic KernelSHAP alternative. The global explanation aggregates feature contributions for a comprehensive view, offering insights into the overall behavior of the AI model. This is particularly important for understanding how the model operates on different traffic datasets (e.g., CIC-DDoS2019,

CICIoT2023, 5 G PFCP). The summary plot (used to represent the global explanation) yields a global view of feature importance across the entire dataset, showing the distribution of SHAP values for each feature and indicating which features have the most significant impact on the AI model's predictions.

The local explanation examines individual data samples, identifying specific feature contributions that influence the AI model's predictions. The force plot (used to represent the local explanation) visualizes the SHAP values for a single data sample, showing how each feature contributes to the prediction. This visualization clearly indicates the features that push the prediction higher or lower relative to the base-line prediction.

3.2.2 LIME

LIME, developed by Ribeiro, Singh, and Guestrin [4], explains complex machine learning models by generating local surrogate models that approximate the AI model's behavior around a specific sample. These surrogate models are simple and interpretable, such as linear regression or decision trees. To obtain an explanation for a specific (original) sample, LIME creates perturbed samples out of the original sample by making slight modifications to its feature values. Each perturbed sample is then inputted to the original AI model to obtain a prediction. In the next step, the perturbed samples are weighted based on their similarity to the original sample, ensuring that perturbed samples that are more similar to the original sample have a greater influence. The weighted perturbed samples and their corresponding predictions are then used to train the surrogate model. The surrogate model approximates the complex original AI model's decision boundary around the specific sample, making it possible to explain the local behavior of the AI model. We encourage readers to refer to the publication [4] for a thorough discussion of the LIME algorithm.

LIME offers several types of visualizations to help users to understand the explanations it generates. One of the primary visualizations is the feature importance plot, which displays the contribution of each feature to the prediction for a specific sample. LIME's model-agnostic nature allows it to be applied to any machine learning algorithm, offering flexibility and versatility. This adaptability makes it particularly useful for local explanations, such as understanding why a specific sample in the CICIoT2023 dataset was classified as malicious. In our framework, LIME is exclusively used in the second stage, as it does not offer global explanations like SHAP. LIME provides local explanations of the AI model's prediction behavior for individual samples, providing end-users with clear insights into the decision-making process.

3.3 Implementation stages

3.3.1 First stage

The process begins with the preprocessing of a well-known offline dataset to ensure the dataset's integrity and relevance while preventing issues such as computational inefficiency and model bias. The cleaned dataset is then normalized using the MinMaxScaler to ensure uniform feature scaling, which is crucial for the convergence and performance of the machine learning model.

Once pre-processed, each dataset is subjected to a stratified 10-fold cross-validation procedure: the shuffled data are partitioned into ten equal folds that preserve the benign/attack class ratio, eight folds ($\approx 80\%$) are used to train the XGBoost model, one fold ($\approx 10\%$) serves as an inner validation set for early stopping and hyper-parameter tuning, and the remaining fold ($\approx 10\%$) provides an unseen test set. This cycle is repeated until every sample has appeared exactly once in the test fold, and we report the mean $\pm$ standard deviation of all performance and explanation-time metrics across the ten runs, yielding reliable estimates with markedly lower variance than a single train/test split. Concurrently, in the feature selection step, ANOVA identifies features with statistically significant differences between classes. The SHAP Global Explanation module complements this by analyzing the model's output to determine the features that are most influential on its predictions.

The Feature Reduction process merges insights from ANOVA and SHAP Global Explanation, assigning a 60% weight to SHAP results and 40% to ANOVA results. This weighting prioritizes SHAP's ability to capture feature importance in the context of model predictions while leveraging ANOVA's statistical rigor to ensure that the selected features are statistically significant. By combining these approaches, we ensure that the most impactful features are retained, balancing computational efficiency with predictive accuracy. Preliminary trials with adjacent ratios (50:50 and 70:30) produced the same top feature list, so 60:40 was kept as a balanced default that modestly favors the model-aware SHAP signal without sacrificing ANOVA's variance-based insight.

The final outcome of the first stage is a reduced set of features that is both statistically significant and crucial to model predictions. These features serve as input for the second stage, reducing dimensionality and computational costs while preserving the essential information needed for high-performance IDS operations.

3.3.2 Second stage

In the second stage, the dataset with reduced features is fed to the XGBoost model for another round of 10-fold cross-

validation. This step evaluates the effectiveness of the feature reduction process by comparing performance metrics before (first stage) and after (second stage) the feature reduction. The expected outcome is that the model sustains high classification performance while requiring fewer features, which translates to reduced computational costs, faster training, and decreased storage needs.

To enhance interpretability, SHAP and LIME Local Explanations are applied to a randomly selected sample to explain the model's decision-making process. SHAP generates explanations by assigning Shapley values to features, showing their contributions to specific predictions. LIME complements this by creating local surrogate models to approximate the decision boundaries of the original model for the selected samples. The synergy between these methods provides a comprehensive view of the feature contributions to each decision.

The explanation consistency check is performed to ensure alignment between SHAP and LIME outputs. This step validates the reliability of the explanations and identifies any inconsistencies or potential biases in feature interpretation. The insights from this process are fed back into the Feature Reassessment loop, where features are re-evaluated, and potentially new features are identified, refining the Feature Reduction process for greater accuracy and efficiency.

This iterative process ensures that our framework adapts dynamically to the characteristics of the dataset, continuously improving both the model's interpretability and prediction performance. The second stage thus serves as the critical link between feature reduction and real-time application, ensuring that the IDS framework remains both efficient and reliable.

4 Performance evaluation

4.1 Datasets, ML model, and performance metrics selection

For evaluating the proposed framework, we select the CIC-DDoS2019 [41], CICIoT2023 [42], and 5 G PFCP [43] datasets, which together cover a broad spectrum of attack traffic and benign flows across different networking environments. CIC-DDoS2019 provides representative volumetric DDoS attacks (e.g., HTTP Flood, SYN Flood) that dominate IP-based networks, CICIoT2023 reflects the complexity of IoT-oriented threats, and 5 G PFCP highlights emerging

control-plane vulnerabilities in mobile-core systems. Within these datasets, we deliberately select a set of representative attacks, namely volumetric (HTTP, SYN), low-and-slow (Slowloris), reconnaissance (Portmap), and 5 G control-plane (Session Deletion, Session Establishment), to capture diverse traffic characteristics and detection challenges. This selection ensures coverage of both common and hard-to-detect attack families, while keeping the evaluation tractable and focused without compromising the generality of the conclusions. Table 3 shows that all three datasets have been pre-processed to be class-balanced but vary in size and number of features. The Portmap set has 59 features for 8000 flows, CICIoT2023 is much larger but reduces to 35 features, and the 5 G PFCP data has just 14 features. This highlights the framework's ability to handle different data scales without being affected by class imbalance. Because the balanced variants of CIC-DDoS2019 (Portmap), the three CICIoT2023 files, and the two 5 G PFCP corpora each contain virtually equal numbers of benign and malicious flows (Table 3), class imbalance was not an issue. Consequently, no resampling or cost-sensitive weighting was applied; instead, all train/validation/test partitions are stratified to preserve the 1:1 class ratio.

Data preprocessing involves three main steps: removing 5-tuple features, removing flow-related features, and removing constant features. The 5-tuple attributes (source IP, destination IP, source port, destination port, and protocol) are removed because they are specific to individual connections. These attributes might also introduce an unnecessary model bias towards specific IP addresses or port numbers, which could make the detection of general attack patterns more difficult. Flow-related features, such as flow duration, are excluded because they require waiting for the entire flow to be completed before being calculated. This step is important for achieving real-time detection, where the IDS must process traffic as it arrives without waiting for flow completion. Constant features are removed because they do not provide any variability or useful information for the model's decision-making process. Since constant features have the same value across all samples, they do not help the model differentiate between malicious and benign traffic and can unnecessarily increase the number of features the model needs to process, potentially leading to increased computational costs without improving performance. After preprocessing, the CIC-DDoS2019 dataset samples have 59 features, the CICIoT2023 dataset samples have 36 features, and the 5 G

Table 3 Summary of the datasets

Dataset	Benign	Attack	Total	Features
CIC-DDoS2019 (Portmap)	4000	4000	8000	59
CICIoT2023 (3 variants)	60,000	60,000	120,000	35
5 G PFCP (2 variants)	3103	3103	6206	14

PFCP dataset samples have 9 features. All features are normalized using the MinMaxScaler to ensure uniformity. While the CIC-DDoS2019 and CICIoT2023 datasets are used for evaluating the XGBoost model's performance before and after feature reduction, the 5 G PFCP dataset is excluded from this step due to its small initial feature set size. Instead, the 5 G PFCP dataset is primarily used to evaluate our framework's ability to explain 5 G control-plane attacks.

The XGBoost model [44] is chosen for the AI-based IDS due to its efficiency, scalability, and ability to handle imbalanced training datasets, along with its reduced computation resource requirements compared to deep learning models. XGBoost's advanced regularization minimizes overfitting, and its gradient boosting framework handles large training datasets effectively. Compared to traditional machine learning models such as Random Forests or Support Vector Machines (SVM), XGBoost requires lower training time and provides better control through hyperparameter tuning. While deep learning models like neural networks can achieve high accuracy, they often require large amounts of training data and computational resources. XGBoost offers a balance by providing good detection performance with lower computational overhead. For datasets like CIC-DDoS2019 and CICIoT2023, this balance is crucial for achieving high performance without excessive computational costs.

To optimize model performance and reduce the risk of overfitting, we employed a hyperparameter tuning procedure using GridSearchCV combined with 10-fold cross-validation. This technique systematically evaluates all possible combinations from a predefined parameter grid, selecting the configuration that yields the best cross-validation performance on the training folds. The full set of candidate values for each parameter is summarized in Table 4, with the final selected settings highlighted in bold. We tuned seven key hyperparameters of the XGBoost classifier. The number of estimators (*n_estimators*) controls the total number of boosting rounds, where a larger number can improve performance but may lead to overfitting if not counterbalanced by regularization. The *learning_rate* governs the contribution of each tree; smaller values require more trees but tend to improve

generalization. The *max_depth* defines the maximum depth of each tree, impacting the model's capacity to capture complex patterns, i.e., higher values increase complexity but may reduce interpretability. The *gamma* parameter imposes a minimum loss reduction required to make a further split, thereby acting as a pruning mechanism. *Reg_lambda* is the L2 regularization term on leaf weights, which penalizes large coefficients to improve model robustness. *Subsample* controls the proportion of training instances randomly sampled for each boosting round, reducing variance and helping avoid overfitting. Similarly, *Colsample_bytree* sets the fraction of features used when building each tree, promoting diversity among trees and lowering the chance of fitting noise. The bolded values in Table 4 indicate the optimal configuration selected by the grid search. These values reflect a balanced trade-off between detection performance and model complexity, confirmed by strong validation results across all datasets. The use of 10-fold stratified cross-validation ensured that the tuning process remained robust to class imbalance and provided reliable estimates of generalization.

To compare the XGBoost model performance before and after feature reduction, we use five metrics: Accuracy, Precision, Specificity, Recall, and F1 Score. These metrics allow for a comprehensive evaluation of our approach's effectiveness beyond mere accuracy. For instance, Precision measures the model's ability to correctly identify malicious samples, while Recall evaluates its capability to detect all malicious samples. Specificity shows the model's ability to avoid false positives, and F1 Score provides a balance between Precision and Recall.

4.2 Evaluation scenarios, results, and analysis

We evaluate our XAI-based IDS framework using one dataset from CIC-DDoS2019 and three datasets from CICIoT2023, each containing benign traffic samples as well as malicious traffic samples of one of the following specific DDoS attack types: Portmap, SYN Flood, HTTP Flood, and Slowloris. These attacks are highly effective and represent both volumetric (e.g., Portmap, SYN Flood, HTTP Flood) and slow (e.g., Slowloris) attack types, which pose significant challenges for an IDS due to their differing traffic patterns. Additionally, we use the 5 G PFCP dataset, which represents control-plane attacks in 5 G networks. It focuses on two specific attack types: PFCP Session Deletion and PFCP Session Establishment. These attacks rely on signaling protocols and packet-level manipulations rather than volumetric traffic patterns seen in data plane attacks. This results in a smaller feature set, often limited to protocol-specific attributes like session counters or message types, which can make it difficult to distinguish malicious activity from normal operations [43].

Table 4 Parameter grid used for XGBoost hyperparameter tuning with GridSearchCV

Parameter	Values
<i>n_estimators</i>	{100, 200, 300 , 400}
<i>learning_rate</i>	{0.05, 0.10 , 0.20}
<i>max_depth</i>	{3, 4, 5 , 6}
<i>gamma</i>	{0, 0.5 , 1}
<i>reg_lambda</i>	{0.5, 1 , 1.5}
<i>subsample</i>	{0.6, 0.8 , 1.0}
<i>colsample_bytree</i>	{0.6, 0.8 , 1.0}

We compare the attack detection results obtained from the first stage of our framework (which utilizes all features, i.e., 59 features for the CIC-DDoS2019 dataset and 36 features for the CICIoT2023 dataset) with those from the second stage after the Feature Reduction. Note that the 5G PFCP dataset is not used for the model performance comparison before and after the Feature Reduction due to its small initial feature set size (9 features). Instead, it is used to evaluate our framework's capability to detect and explain 5G-specific control-plane attacks using both global and local explanations. The Feature Reduction process, based on SHAP Global Explanation and statistical methods, selects features that are both statistically significant and practically impactful, improving the framework's efficiency without sacrificing prediction accuracy. To assess the effect of Feature Reduction, we examine the XGBoost model performance by applying the top 5, 10, 20, and 30 features. This evaluation enables us to understand the trade-offs between reducing computational overhead and maintaining high detection performance, especially for datasets with a larger number of initial features.

4.2.1 Results of SHAP global explanation, feature selection, and feature reduction processes

Figures 2, 3, and 4 display the SHAP global summary plots for the Portmap, SYN Flood, HTTP Flood, Slowloris,

PFCP Session Deletion, and PFCP Session Establishment attacks, respectively, highlighting the most influential features on model predictions. The Y-axis lists the features in the order of their importance, while the X-axis, indicating the SHAP values, shows each feature's impact on the model's prediction. Positive SHAP values push predictions towards the malicious class, while negative values push towards the benign class. The explanations for two distinct DDoS attack types: volumetric (Portmap, SYN Flood, HTTP Flood) and slow (Slowloris) are investigated. For the Portmap attack, the 'Fwd Packet Length Min' feature emerges as the most significant. Lower values of this feature, indicated by blue dots, significantly decrease the model's output and push the prediction towards the benign class. The 'Inbound' feature is the second most important one, with higher values (red dots) generally pushing the prediction towards the malicious class. Similarly, for the SYN Flood attack, 'syn_flag_number' and 'Number' emerge as the most significant features, with high values of these features strongly indicating SYN Flood attack samples. The 'Header_Length' feature is the next most important one, where lower values are indicative of SYN Flood attacks. For the HTTP Flood attack, the 'Number' feature, representing the number of packets in a flow, is the most influential one. The 'fin_count' feature, which counts packets with the FIN flag set, also plays a crucial role. Higher values tend to push predictions towards

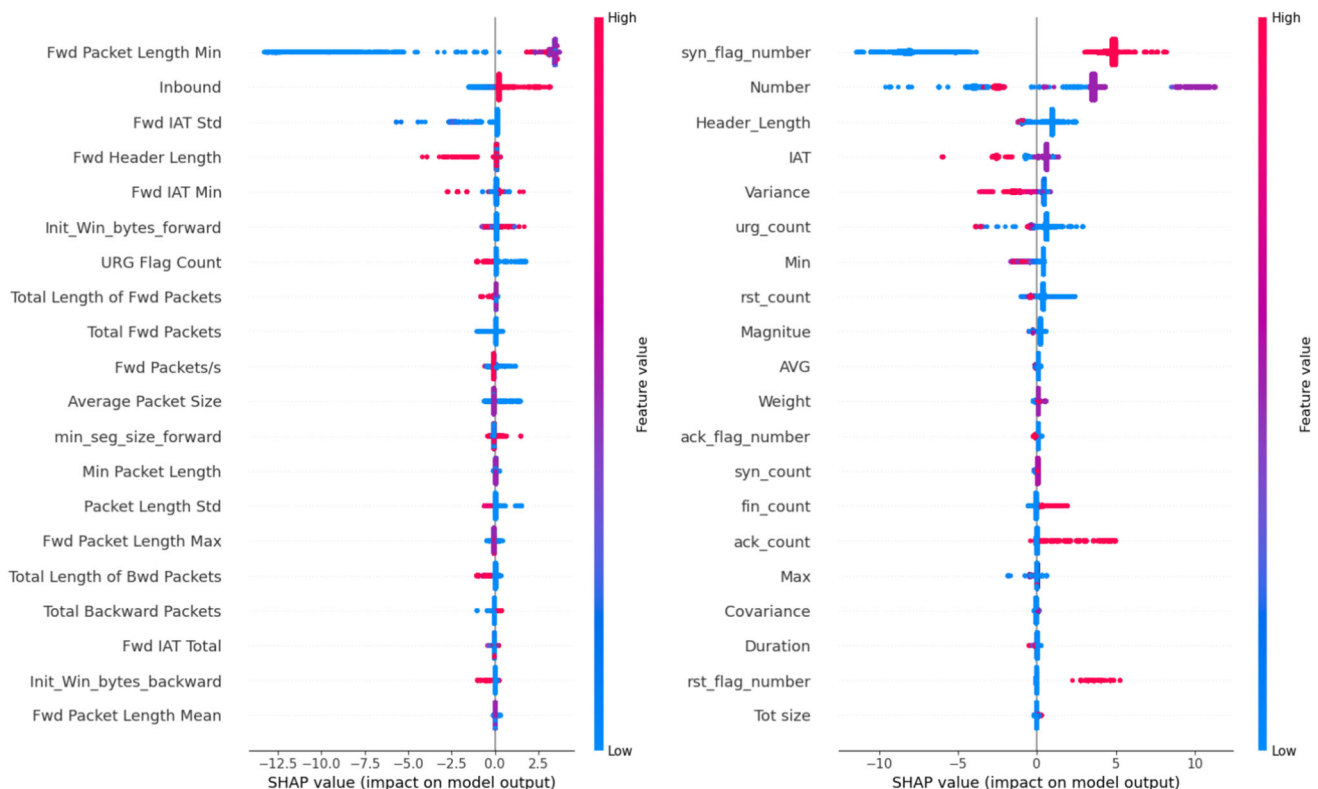


Fig. 2 SHAP global explanation results for Portmap and SYN Flood attacks

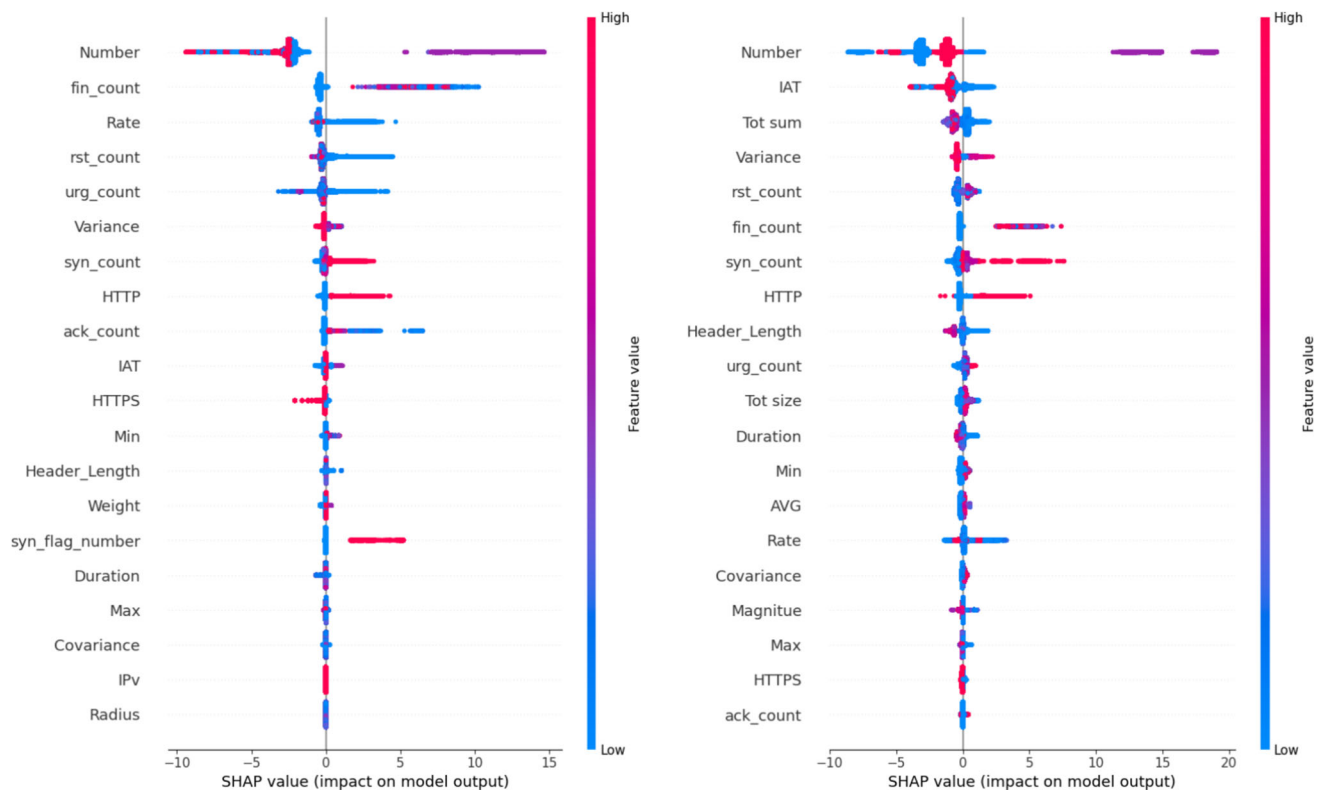


Fig. 3 SHAP global explanation results for HTTP Flood and Slowloris attacks

the malicious class. Similarly, in the case of Slowloris, the 'Number' feature remains the most important one. Higher values significantly increase the model's output, driving predictions towards the malicious class. The 'IAT' feature, representing the packet inter-arrival time, is also highly influential, with lower values pushing predictions towards a malicious outcome. In the case of 5G control-plane attacks, the framework demonstrates its capability to identify control-plane attack-specific features. For the PFCEP Session Deletion attack, 'PFCEPHeartbeatRequest_counter' is the most influential feature. Low values of 'PFCEPHeartbeatRequest_counter' are indicative of malicious samples,

highlighting the role of this feature in identifying abnormalities in PFCEP Heartbeat requests. Similarly, for the PFCEP Session Establishment attack, 'PFCEPSessionEstablishmentRequest_counter' emerges as the most crucial feature, with low values strongly linked to predictions of malicious traffic. These results show the importance of control-plane-specific features in detecting such threats.

To complement the results of the SHAP Global Explanation, we utilize the ANOVA feature selection method, which identifies statistically significant features by analyzing the variance between classes. ANOVA, or Analysis of Variance, evaluates the degree to which individual features

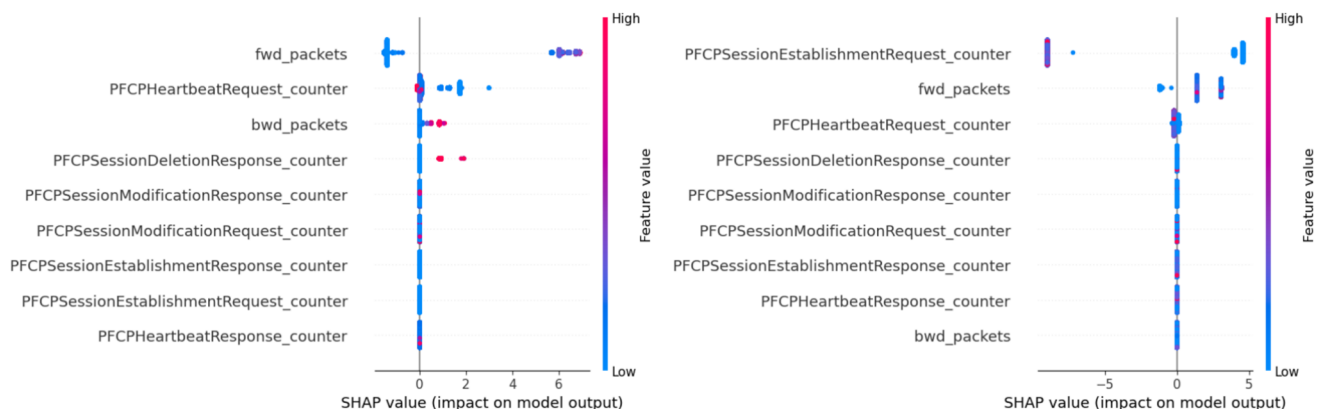


Fig. 4 SHAP global explanation results for PFCEP attacks

contribute to class separation by comparing the variance within classes (e.g., benign or malicious traffic) to the variance between classes. Features with a higher between-class variance than within-class variance are considered statistically significant, making them crucial for distinguishing between malicious and benign traffic. ANOVA is particularly advantageous due to its simplicity, efficiency, and ability to handle large datasets. Unlike more complex feature selection techniques, such as Recursive Feature Elimination (RFE), Mutual Information (MI), or Wrapper Methods, ANOVA does not rely on model-specific parameters and offers a purely statistical perspective, making it a robust complement to model-based interpretability tools like SHAP. By combining the outcomes of SHAP Global Explanation and ANOVA, we achieve a more comprehensive selection of features, integrating SHAP's interpretability with ANOVA's statistical precision. The Feature Reduction process merges these

results, assigning a 60% weight to SHAP Global Explanation and 40% to ANOVA, prioritizing SHAP's ability to capture feature importance in the context of model predictions while incorporating the statistical insights from ANOVA. Table 5 presents the top 30 features for the CICIoT2023 and CIC-DDoS2019 datasets, and Table 6 lists the 9 features for the 5 G PCFP datasets. These identified features align well with the nature of each attack. For instance, in volumetric attacks, which rely on generating a high rate of traffic to overwhelm the target, traffic volume-related features such as 'Number' are among the most important. On the other hand, for the Slowloris attack, 'HTTP' is a key feature, surpassed only by the 'Number' feature, as this attack relies on slowly sending incomplete HTTP requests to keep connections open. These tables represent the final results of the first stage and serve as the input for further analysis and evaluation in the second stage of our framework.

Table 5 Most important features as identified in the first stage

Portmap	SYN Flood	HTTP Flood	Slowloris
Fwd Packet Length Min	syn_flag_number	Number	Number
Min Packet Length	Number	HTTP	HTTP
Inbound	rst_count	urg_count	syn_count
Avg Fwd Segment Size	IAT	HTTPS	IAT
Fwd Packet Length Mean	Weight	rst_count	rst_count
URG Flag Count	Magnitude	Duration	Tot sum
Average Packet Size	Header_Length	syn_count	Header_Length
Fwd Packets/s	Variance	ack_count	Variance
Fwd Header Length	Min	fin_count	Duration
Fwd IAT Total	ack_flag_number	Header_Length	HTTPS
Packet Length Mean	fin_count	Magnitude	fin_count
Packet Length Std	urg_count	Rate	rst_flag_number
Fwd IAT Std	Tot sum	AVG	Tot size
Down/Up Ratio	ack_count	Tot size	psh_flag_number
Bwd IAT Mean	HTTPS	Std	AVG
Init_Win_bytes_forward	Duration	Radius	urg_count
Fwd IAT Min	syn_count	Max	Magnitude
Total Length of Fwd Packets	Std	Tot sum	ack_count
Total Length of Bwd Packets	Radius	Variance	Rate
Init_Win_bytes_backward	Covariance	Min	Max
Bwd Packet Length Min	AVG	syn_flag_number	Std
Bwd IAT Min	Tot size	IAT	Min
CWE Flag Count	Max	TCP	Radius
ACK Flag Count	rst_flag_number	Covariance	Srate
Bwd Packet Length Mean	UDP	rst_flag_number	fin_flag_number
Avg Bwd Segment Size	TCP	UDP	Covariance
Bwd IAT Total	HTTP	Weight	TCP
Bwd IAT Std	psh_flag_number	psh_flag_number	ack_flag_number
Bwd Packet Length Max	Rate	Srate	UDP
Fwd PSH Flags	DNS	fin_flag_number	DNS

Table 6 Most important features as identified in the first stage (continued)

PFCP Session Deletion	PFCP Session Establishment
fwd_packets	PFCPSessionEstablishmentRequest_counter
PFCPSessionModificationRequest_counter	PFCPSessionEstablishmentResponse_counter
PFCPSessionModificationResponse_counter	PFCPHeartbeatRequest_counter
PFCPHeartbeatRequest_counter	PFCPHeartbeatResponse_counter
bwd_packets	fwd_packets
PFCPSessionDeletionResponse_counter	bwd_packets
PFCPHeartbeatResponse_counter	PFCPSessionModificationRequest_counter
PFCPSessionEstablishmentRequest_counter	PFCPSessionModificationResponse_counter
PFCPSessionEstablishmentResponse_counter	PFCPSessionDeletionResponse_counter

Table 7 Performance metric comparison for Portmap

	Accuracy	Precision	Specificity	Recall	F1 Score
All Features	99.813%	99.874%	99.876%	99.749%	99.811%
Top 5	81.063%	77.666%	75.249%	86.935%	82.039%
Top 10	98.125%	97.875%	97.886%	98.367%	98.120%
Top 20	98.500%	98.371%	98.383%	98.618%	98.494%
Top 30	98.875%	98.869%	98.881%	98.869%	98.869%

Table 8 Performance metric comparison for SYN Flood

	Accuracy	Precision	Specificity	Recall	F1 Score
All Features	99.988%	100.000%	100.000%	99.975%	99.988%
Top 5	96.438%	96.207%	96.114%	96.755%	96.480%
Top 10	99.250%	99.135%	99.117%	99.381%	99.258%
Top 20	99.788%	99.777%	99.773%	99.802%	99.789%
Top 30	99.950%	99.950%	99.950%	99.950%	99.950%

Table 9 Performance metric comparison for HTTP Flood

	Accuracy	Precision	Specificity	Recall	F1 Score
All Features	100.000%	100.000%	100.000%	100.000%	100.000%
Top 5	96.281%	96.258%	96.035%	96.514%	96.386%
Top 10	99.513%	99.633%	99.616%	99.413%	99.523%
Top 20	99.688%	99.731%	99.719%	99.658%	99.694%
Top 30	99.888%	99.878%	99.872%	99.902%	99.890%

Table 10 Performance metric comparison for Slowloris

	Accuracy	Precision	Specificity	Recall	F1 Score
All Features	99.988%	100.000%	100.000%	99.975%	99.987%
Top 5	97.013%	98.077%	98.130%	95.890%	96.971%
Top 10	99.013%	99.121%	99.127%	98.897%	99.009%
Top 20	99.725%	99.724%	99.726%	99.724%	99.724%
Top 30	99.925%	99.925%	99.925%	99.925%	99.925%

4.2.2 Performance metrics comparison: Before vs after feature reduction

Tables 7, 8, 9, and 10 give a detailed comparison of the performance metrics of the XGBoost model in the first stage (all features) and the second stage (after feature reduction) for the CIC-DDoS2019 and CICIoT2023 datasets, thereby quantifying the impact of feature reduction on detection quality. Figure 5 complements these tables by visualizing how Accuracy and F1 evolve with $K \in \{5, 10, 20, 30, \text{all}\}$: performance remains flat from “All Features” down to the Top-10 set and degrades noticeably only at Top-5, confirming that ten features constitute a robust operating point. Across the ten stratified cross-validation folds, the average absolute gap between validation and test accuracies is below 0.10 percentage points for every attack class (Portmap: 0.09 pp, SYN Flood: 0.02 pp, HTTP Flood: 0.01 pp, Slowloris: 0.03 pp; standard deviations range from 0.01 pp to 0.13 pp). The average absolute gap between validation and test accuracies is defined as the mean of $|\text{Acc}_{\text{val},i} - \text{Acc}_{\text{test},i}| \times 100\%$ over all ten folds, i.e. the percentage-point difference between the accuracy measured on the inner validation subset and that on the corresponding held-out test subset, so smaller gaps

Table 11 Confusion matrix - Portmap (Top-10 features, CIC-DDoS2019)

	Pred. Neg.	Pred. Pos.
Actual Neg.	787 (TN)	17 (FP)
Actual Pos.	13 (FN)	783 (TP)

indicate closer agreement and stronger generalization. These results confirm that the model generalizes extremely well and that the metrics reported in the tables are reliable.

For the Portmap attack, retaining the top 10 features sustains a high accuracy of 98.125% and precision of 97.875%; a further reduction to 5 features causes a marked decline (81.063% accuracy, 77.666% precision), indicating loss of critical information.

For SYN Flood, performance remains strong with 10 features (99.250% accuracy, 99.381% recall), whereas reducing to 5 features lowers accuracy to 96.438% and recall to 96.755%. A similar trend is observed for HTTP Flood: accuracy stays above 99.513% with 10 features but recall and F1 Score fall to 96.514% and 96.386% with 5 features.

Accuracy and F1 Score Trends Across Feature Sets for Each Attack

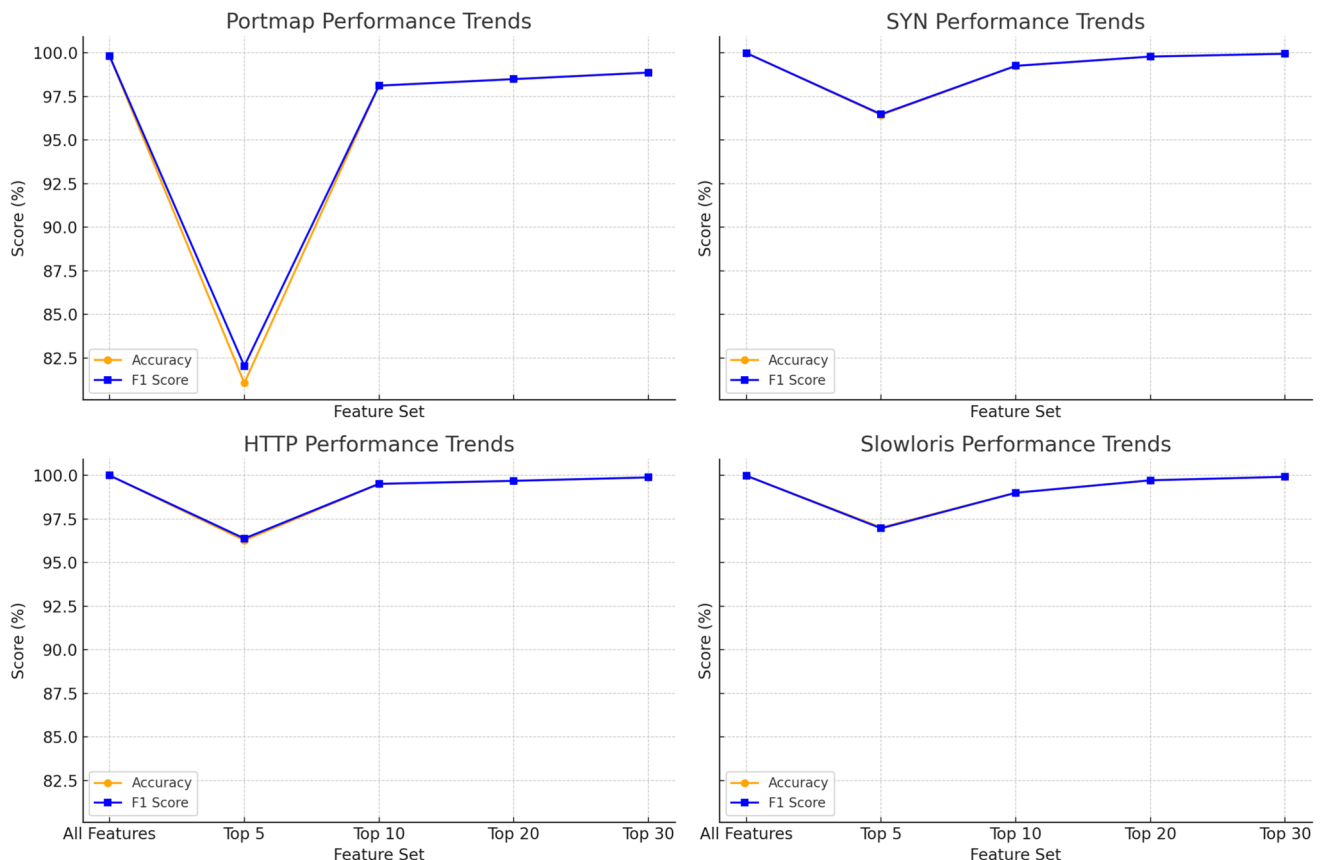


Fig. 5 Performance trends across feature sets

Slowloris detection remains excellent with 20 features; trimming to 10 features causes a slight dip (99.013% accuracy, 99.121% precision), and to 5 features lowers recall and F1 Score to 95.890% and 96.971%, respectively, reflecting the subtler traffic signatures of slow-rate attacks. The confusion-matrix tables (Tables 11, 12, 13, and 14) show the detection performance of the best Top-10 feature configurations.

Table 15 complements the XGBoost results in Table 9 with two classical baselines. Random Forest attains virtually perfect metrics under both feature settings (only a 0.0001 loss in F1 Score when the dimensionality is reduced), which is expected because bootstrap aggregation and per-split feature subsampling already mitigate the impact of redundant predictors. Linear SVM, on the other hand, scores markedly lower when trained on all features (F1 Score to be around 79%) but rises to 94% after the Top-10 reduction; eliminating irrelevant or weakly informative dimensions increases the margin separability of the classes and thereby improves its generalization. The combined evidence indicates that the ANOVA+SHAP filter retains the discriminative power needed for high detection accuracy while simplifying the input space—benefiting margin-based learners such as SVM—whereas our two-stage XGBoost pipeline preserves Random Forest-level performance and, as shown in Sect. 4.2.4, yields the fastest and most reliable local explanations.

The feature reduction process is crucial for balancing performance and efficiency, ensuring that the proposed framework remains scalable while preserving its detection capabilities. In the subsequent sections, we explore how this reduction affects explanation generation time, investigating its role in enhancing the practical applicability of the framework.

4.2.3 SHAP and LIME local explanations

The results of the SHAP Local Explanations are visualized in the force plots, which illustrate how individual features contribute to the XGBoost model's specific prediction. As shown in Figs. 6, 7, and 8, the plots begin at a base value that represents the average prediction result across all samples of the respective dataset, denoted by a gray vertical line. Features that push the prediction higher (indicating malicious traffic) are marked in red, while those that push it lower (indicating benign traffic) are marked in blue. The bar lengths corre-

Table 13 Confusion matrix - HTTP Flood (Top-10 features)

	Pred. Neg.	Pred. Pos.
Actual Neg.	3,894 (TN)	15 (FP)
Actual Pos.	24 (FN)	4,067 (TP)

spond to the magnitude of each feature's impact. The final prediction is computed by adding these contributions to the base value.

As an example, first we examine the SHAP force plots for SYN Flood and Slowloris attack samples, depicted in Figs. 6 to 7. In the SYN Flood plot, the base value starts at 3.36, reflecting the model's initial prediction before incorporating feature contributions. The final output value of 15.64 is achieved primarily due to the substantial contributions of features such as 'syn_flag_number' and 'Number', which strongly push the prediction toward the malicious class. Similarly, in the Slowloris plot, the base value starts at -0.06675, indicating a baseline heavily leaning towards a benign outcome. The final output value increases sharply to 9.94, primarily driven by the 'Number' and 'fin_count' features, which have significant positive contributions towards the malicious class. This pattern underscores the model's capability to identify slow DDoS attacks like Slowloris that rely on specific traffic patterns. Although less prominent in the plot, features such as 'syn_count' and 'IAT' also exhibit small positive SHAP values, contributing to the overall prediction.

Second, we observe the SHAP force plots for the 5 G PFCP attack samples, depicted in Fig. 8. For the PFCP Session Deletion attack, the base value starts at 0.1034 and rises to 7.87, primarily driven by the features 'fwd_packets' and 'PFCPHeartbeatRequest_counter'. Similarly, for the PFCP Session Establishment attack, the base value starts at -0.0606 and increases to 2.38, with significant contributions from the 'PFCPSessionEstablishmentRequest_counter' feature.

Next, we consider the LIME feature importance plots shown in Figs. 9, 10, and 11, which illustrate how individual features influence specific predictions made by the XGBoost model. The prediction probabilities for the negative (benign) and positive (malicious) classes are displayed at the top left of each plot, providing insight into the model's confidence in its prediction. The features are listed alongside their values, and their contributions are visually represented in orange or

Table 12 Confusion matrix - SYN Flood (Top-10 features)

	Pred. Neg.	Pred. Pos.
Actual Neg.	3,928 (TN)	35 (FP)
Actual Pos.	25 (FN)	4,012 (TP)

Table 14 Confusion matrix - Slowloris (Top-10 features)

	Pred. Neg.	Pred. Pos.
Actual Neg.	3,975 (TN)	35 (FP)
Actual Pos.	44 (FN)	3,946 (TP)

Table 15 Classical-ML baselines on HTTP Flood traffic: full feature set vs. Top-10 ANOVA+SHAP features

Model	Feature set	Accuracy	Recall	F1
Random Forest	All Features	0.9999	0.9998	0.9999
Random Forest	Top 10	0.9998	0.9995	0.9998
[2pt] Linear SVM	All Features	0.7833	0.7952	0.7896
Linear SVM	Top 10	0.9329	0.9768	0.9370

blue bars. Orange indicates features pushing the prediction towards the malicious class, while blue represents features driving the prediction towards the benign class. The length of each bar corresponds to the magnitude of the feature's contribution, with longer bars indicating a larger impact on the prediction outcome.

For the SYN Flood attack, the LIME plot identifies the feature 'Number' with a value of 9.50 as the primary contributor to the malicious prediction, accounting for a significant portion of the probability (0.95). Additional features such as 'psh_flag_number' and 'ack_count' also contribute to the malicious classification, reinforcing the prediction. In contrast, features such as 'DNS' and 'urg_count' push the prediction towards the benign class but have a smaller overall influence.

For the Slowloris attack, the feature 'Number' again turns out to be most significant, followed by the features 'fin_count', 'DNS', and 'syn_count'. These features contribute predominantly to the malicious class, emphasizing the model's reliance on traffic volume and flag-related features to detect Slowloris behavior effectively. Features such as 'IAT' and 'ack_flag_number' exhibit smaller contributions, which aligns with the subtle nature of Slowloris traffic patterns.

For the 5 G PFCP attacks, the LIME plots provide valuable insights into the control-plane traffic features that influence the predictions. For the PFCP Session Deletion attack, features such as 'fwd_packets' and 'PFCPHeartbeatRequest_counter' are significant contributors to the malicious classification. The feature 'fwd_packets' with a high value (506) pushes strongly towards the malicious class, while the feature 'PFCPHeartbeatRequest_counter' also plays an important role in the decision-

making process. Similarly, for the PFCP Session Establishment attack, the features 'PFCPSessionEstablishmentRequest_counter' and 'fwd_packets' dominate the prediction. In particular, the feature 'PFCPSessionEstablishmentRequest_counter' with a low value (0.00) strongly pushes the prediction towards the malicious class.

Our framework includes an explanation consistency check process to ensure the alignment of insights gained from SHAP and LIME local explanations, which enhances the reliability and trustworthiness of the model's interpretations. By comparing the explanations generated by both methods on the same sample, we can identify consistencies and discrepancies in feature importance, which helps to validate the model's decisions. This cross-verification is particularly valuable in scenarios where interpretability and accountability are critical, such as in detecting complex attack patterns.

For example, in the case of a sample of a SYN Flood attack, both SHAP and LIME consistently identified the 'Number' feature as the most influential factor driving the prediction toward the malicious class. This consistency underscores the robustness of the framework's ability to detect SYN Flood attacks with high confidence. However, discrepancies are also observed: For example, SHAP identified the features 'IAT' and 'Header_Length' as crucial in pushing the prediction towards the malicious outcome, whereas these features are less emphasized by LIME. Conversely, LIME, contrary to SHAP, emphasizes the feature 'psh_flag_number' as an important contributor. These differences likely arise due to the inherent methodological variations in how SHAP and LIME generate explanations. SHAP considers all possible feature interactions in its game-theoretic approach, while LIME relies on perturbations and surrogate models to approximate feature influence locally. Understanding these methodological differences can help to refine the feature selection process and to improve the model interpretability further.

For a sample of a Slowloris attack, both SHAP and LIME consistently identified the features 'Number' and 'fin_count' as significant contributors to the positive classification, confirming their importance. This consistency indicates that both methods effectively capture the crucial

**Fig. 6** SHAP local explanation results for Portmap and SYN Flood attacks

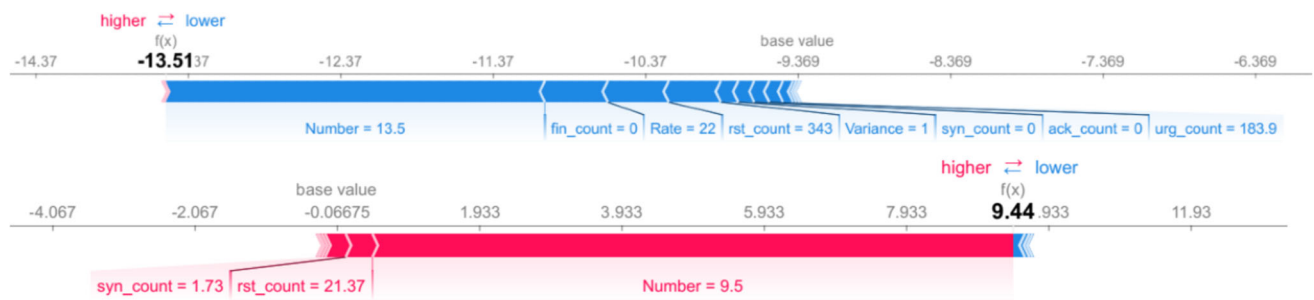


Fig. 7 SHAP local explanation results for HTTP Flood and Slowloris attacks

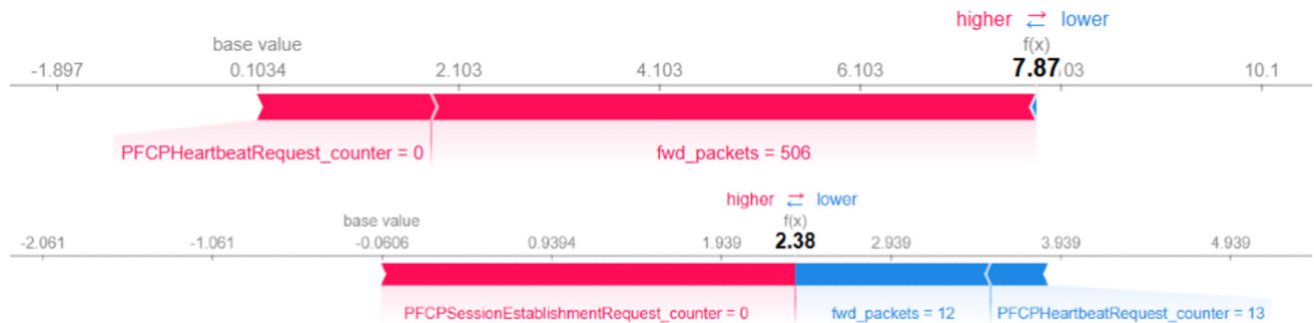


Fig. 8 SHAP local explanation results for PFCP Session Deletion and Session Establishment attacks



Fig. 9 LIME local explanation results for Portmap and SYN Flood attacks

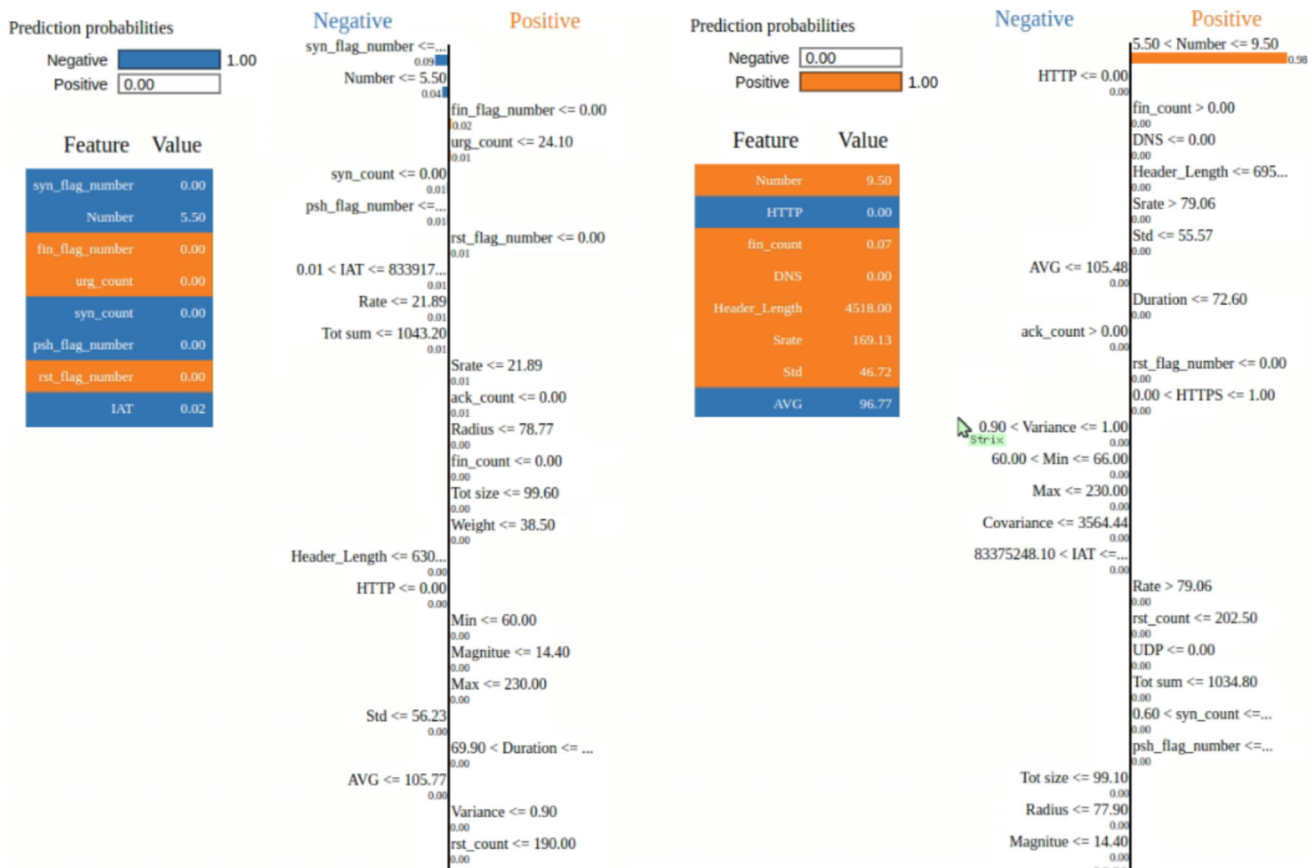


Fig. 10 LIME local explanation results for HTTP Flood and Slowloris attacks

features needed to detect slow DDoS attacks. Nonetheless, minor discrepancies were observed in the ranking and magnitude of other features, e.g., ‘syn_count’ and ‘DNS’, which appeared more prominently in LIME compared to SHAP. These variations might reflect differences in how each method prioritizes subtle feature contributions in low-intensity attack traffic scenarios like Slowloris attacks.

These differences indicate areas where further analysis or a feature reassessment might be required to enhance the interpretability and ensure the robustness of the detection model. This might involve analyzing the feature impact for different samples, exploring feature engineering techniques, and adjusting the model hyperparameters. For instance, features that show a consistent importance for attack detection in both methods could be further validated by using different samples that are classified as malicious (by XGBoost) to assess their generalizability. Similarly, discrepancies between SHAP and LIME could indicate opportunities to improve feature selection or to refine the hyperparameters in the training process.

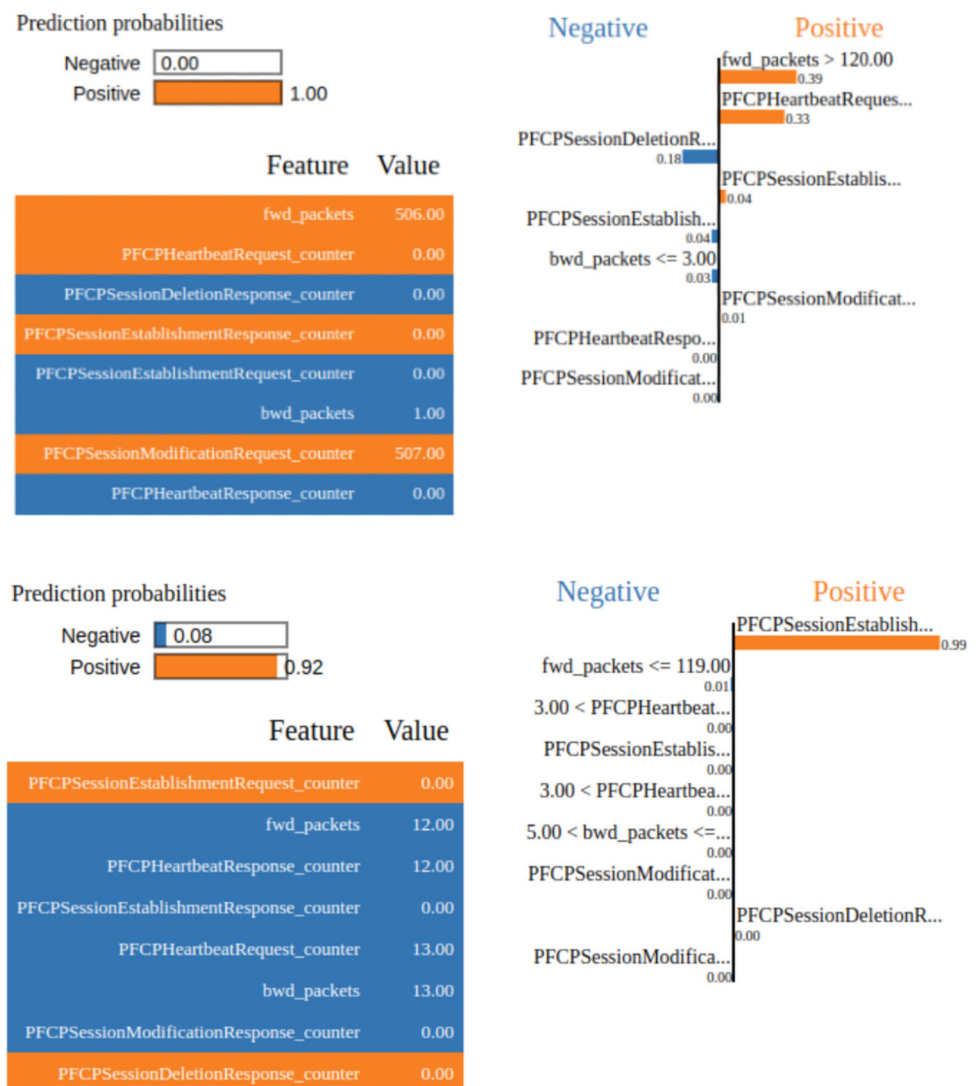
With this iterative process, it is ensured that the most significant features are retained, enhancing the overall IDS detection performance and interpretability. Furthermore, incorporating additional consistency checks for different attack types and networking environments, such as IoT or

5G, could further solidify the framework’s reliability. This approach not only strengthens trust in the IDS’s decision-making but also supports its adaptability to evolving cyber threats.

4.2.4 Efficiency of the local explanations

In this subsection, we evaluate the efficiency of SHAP and LIME local explanations by comparing their execution times for two scenarios: using all features and using the reduced feature set obtained through the first stage of our framework. All experiments were conducted on a workstation with an AMD Ryzen 7 3800X 8-core CPU, 32 GB RAM, and an NVIDIA RTX 3060 GPU (12 GB VRAM). For LIME, we used a single CPU core with 5000 perturbations per instance to provide consistent timing comparisons. While the reported measurements reflect this specific hardware configuration, the framework’s design—particularly the dimensionality reduction stage and the use of a lightweight XGBoost model—ensures that detection accuracy is preserved even on lower-specification systems. The primary difference in such environments would be in explanation generation time, especially for LIME, which is more sensitive to CPU performance than SHAP’s TreeExplainer.

Fig. 11 LIME local explanation results for PFCP Session Deletion and Session Establishment attacks



The results are illustrated in Fig. 12.

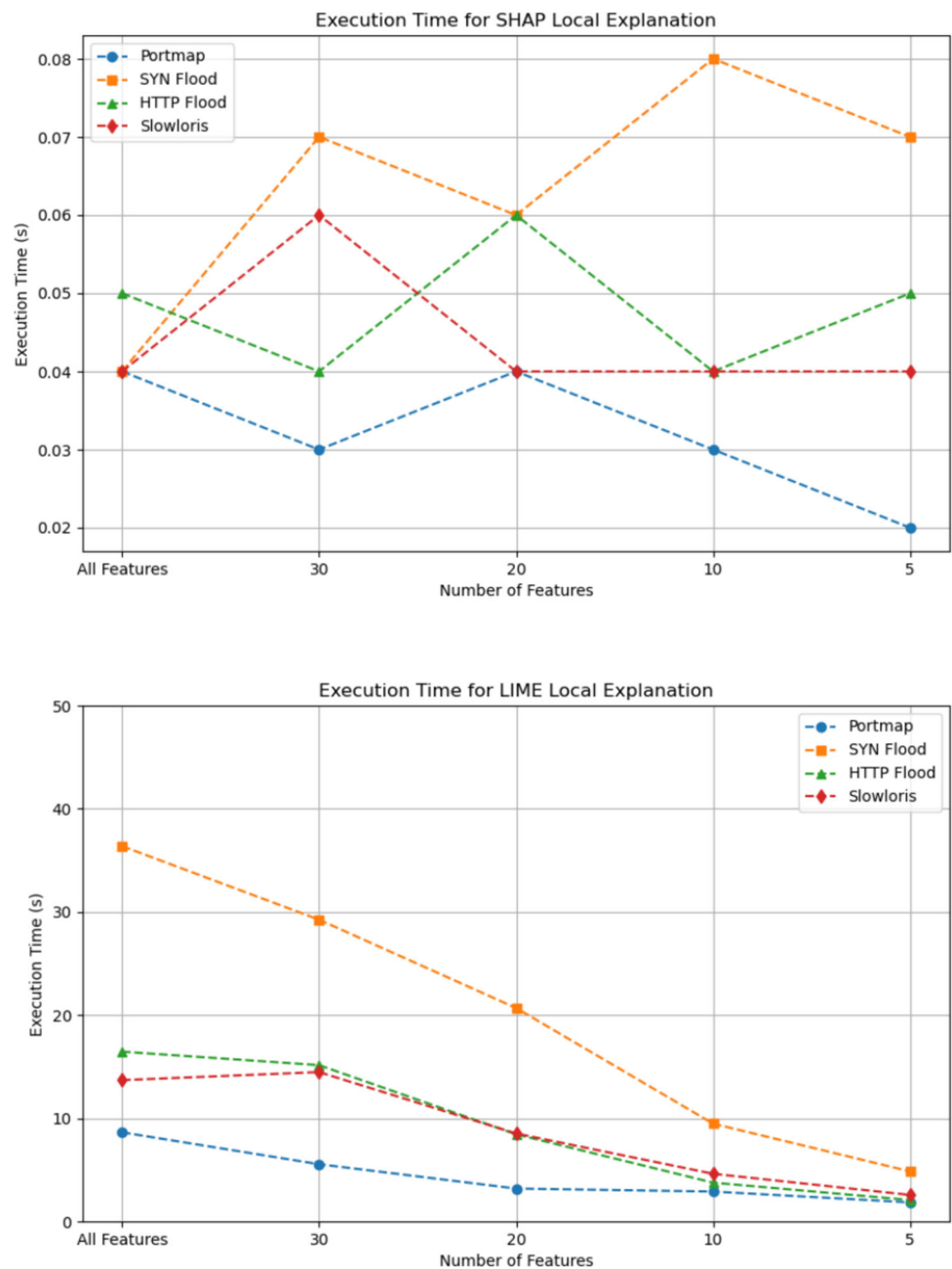
As observed, SHAP Local Explanation demonstrates consistently high efficiency across all datasets, with execution times remaining low, ranging between 0.02 and 0.08 s, irrespective of the number of features. This efficiency makes SHAP particularly suitable for real-time IDS applications where rapid insights are important.

In contrast, LIME Local Explanation shows a significant dependency on the number of features—the execution times dramatically decrease when the feature set is reduced. For instance, in the case of the SYN Flood, the execution time drops from 36.38 s when using all features to 9.48 s with 10 features, representing a 74% time reduction. Further reducing the feature set to 5 features decreases the execution time to 4.85 s, achieving an 87% speed-up compared to using all features. A similar behavior is observed for other attack traffic datasets, showing the substantial improvement in LIME's efficiency brought about by feature reduction. However, as

demonstrated in Sect. 4.2.2, a reduction to only 5 features leads to a significant reduction in detection performance. The 10-feature set turns out to be the optimal choice as it strikes a balance between speed and accuracy, which is essential for real-time IDS applications where both computational efficiency and prediction quality are crucial.

For the next evaluation, we combined our proposed two-stage method with the attention weights-based approach from Santhiappan et al. [33], originally designed for text data using the CAML architecture to enhance LIME's efficiency by emphasizing relevant features. We adapted their method to tabular datasets (CIC-DDoS2019 and CICIOT2023) by utilizing the TabNet architecture [45], a deep learning model specifically designed for tabular data instead of the CAML architecture. TabNet incorporates a sparse attention mechanism that dynamically selects the most relevant features for each instance, mimicking decision tree-like interpretability while leveraging the flexibility and power of neural

Fig. 12 Execution time for the SHAP and LIME local explanations with XGBoost model

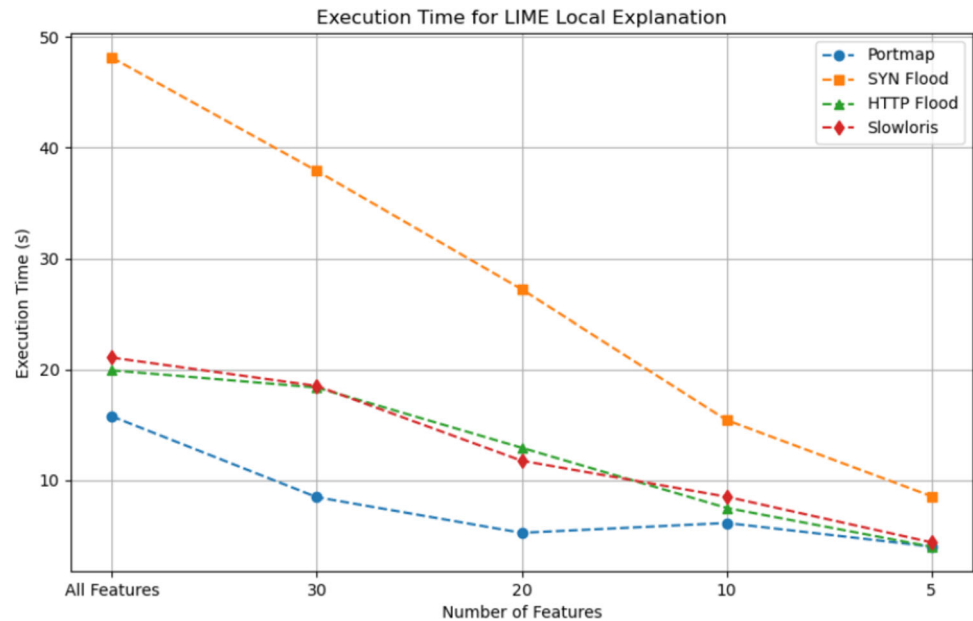


networks. Its Feature Transformer processes input features to capture complex relationships, while the iterative decision steps refine the predictions by progressively building hierarchical feature representations. We fine-tuned the hyper-parameters of the TabNet architecture using GridSearchCV. The resulting configuration is as follows: the model uses eight decision dimensions ($n_d=8$) and eight attention dimensions ($n_a=8$), with three decision steps ($n_{steps}=3$) to iteratively refine feature importance and predictions. The learning rate ($lr=0.02$) ensures steady optimization, and batch processing is optimized with a batch size of 512 and a virtual batch size of 128 to balance memory efficiency and training

speed. These parameters, combined with the sparse attention mechanism, allow TabNet to dynamically focus on significant features in the datasets.

Furthermore, we exchanged the XGBoost model in our framework by the method proposed by Santhiappan et al. [33], which employs the Convolutional Attention for Multi-Label classification (CAML) model, an attention weight-based model designed for text data. Their method enhances LIME's efficiency by using attention weights derived from CAML to focus on the most relevant features. This approach reduces the number of features that LIME needs to examine, thereby accelerating the explanation generation time. Since

Fig. 13 Execution time for LIME local explanations with TabNet model



CAML is unsuitable for tabular datasets and XGBoost is not an attention weights-based model, we replaced CAML with TabNet [45], which is a deep learning model specifically designed for tabular data and is also equipped with an attention mechanism. TabNet uses a sparse attention mechanism that dynamically selects the most relevant features for each sample during training. The attention-based Feature Transformer focuses on important features and learns their significance. Through an iterative process, TabNet refines its predictions step by step, with each step building on the knowledge gained from earlier steps to improve its understanding of the data. We fine-tuned the hyperparameters of the TabNet architecture using GridSearchCV. The resulting configuration is as follows: the model uses eight decision dimensions ($n_d=8$) and eight attention dimensions ($n_a=8$), with three decision steps ($n_{steps}=3$) to iteratively refine feature importance and predictions. The learning rate ($lr=0.02$) ensures steady optimization, and batch processing is optimized with a batch size of 512 and a virtual batch size of 128 to balance memory efficiency and training speed. These parameters, combined with the sparse attention mechanism, allow TabNet to dynamically focus on significant features in the datasets.

We conducted experiments on CIC-DDoS2019 and CICIoT2023 datasets, and analyzed the explanation generation time, focusing on the impact of feature reduction. As shown in Fig. 13, the explanation generation time decreases significantly when fewer features are used. For instance, in the SYN Flood dataset, the time dropped from 48.13 to 15.43 s when using 10 features, and for Portmap, it decreased from 15.77 to 6.14 s. These results confirm that the Feature Reduction process introduced in our framework improves the efficiency of LIME's explanation generation when paired

with attention-weight-based models like TabNet. In addition, we observe that using XGBoost achieves faster explanation generation times, as TabNet's attention mechanism introduces additional computational complexity. The codes and datasets utilized in this study are publicly available online¹ to facilitate full reproducibility.

4.2.5 Ablation study

Table 16 shows the ablation results for Portmap traffic, where both SHAP-only and ANOVA+SHAP fusion maintain perfect alignment with the full-feature baseline ($F1 = 0.9994$). ANOVA-only selection, however, causes a noticeable decline ($F1 = 0.9981$), indicating that purely statistical filtering may miss interaction effects critical to Portmap classification. From an efficiency perspective, SHAP-only achieves the lowest LIME time (1.85 s), while the combined variant retains the same detection quality with only a marginal increase (2.29 s), suggesting that fusion preserves interpretability without introducing overhead.

For SYN Flood (Table 17), all models achieve near-identical performance to the full-feature baseline ($F1 = 0.9999$), confirming the separability of this attack class. ANOVA-only again exhibits a minor drop ($F1 = 0.9990$), but it reduces LIME explanation time to 1.90 s, while SHAP-only requires 2.18 s. The ANOVA+SHAP fusion offers a balanced compromise (2.29 s), maintaining perfect accuracy while trimming redundant features.

The HTTP Flood subset (Table 18) further illustrates these trends. Both SHAP-only and ANOVA+SHAP achieve perfect scores, while ANOVA-only shows a small reduction ($F1$

¹ <https://github.com/DLTeamTUC/XAI-IDS>

Table 16 Ablation study on Portmap traffic. Top-10 variants are obtained with (i) ANOVA only, (ii) SHAP only, and (iii) the proposed ANOVA + SHAP fusion

Variant	Accuracy	Recall	F1	LIME time (s)
XGBoost (All features)	0.9994	0.9987	0.9994	–
[2pt] XGBoost + ANOVA Top-10	0.9981	0.9962	0.9981	2.32
XGBoost + SHAP Top-10	0.9994	0.9987	0.9994	1.85
XGBoost + ANOVA + SHAP Top-10	0.9994	0.9987	0.9994	2.29

Table 17 Ablation study on SYN Flood traffic. Top-10 variants are obtained with (i) ANOVA only, (ii) SHAP only, and (iii) the proposed ANOVA + SHAP fusion

Variant	Accuracy	Recall	F1	LIME time (s)
XGBoost (All features)	0.9999	0.9998	0.9999	–
[2pt] XGBoost + ANOVA Top-10	0.9990	0.9980	0.9990	1.90
XGBoost + SHAP Top-10	0.9999	0.9998	0.9999	2.18
XGBoost + ANOVA + SHAP Top-10	0.9999	0.9998	0.9999	2.29

Table 18 Ablation study on HTTP Flood traffic. Top-10 variants are obtained with (i) ANOVA only, (ii) SHAP only, and (iii) the proposed ANOVA + SHAP fusion

Variant	Accuracy	Recall	F1	LIME time (s)
XGBoost (All features)	1.0000	1.0000	1.0000	–
[2pt] XGBoost + ANOVA Top-10	0.9969	0.9952	0.9969	2.33
XGBoost + SHAP Top-10	1.0000	1.0000	1.0000	2.30
XGBoost + ANOVA + SHAP Top-10	1.0000	1.0000	1.0000	2.16

Table 19 Ablation study on Slowloris traffic. Top-10 variants are obtained with (i) ANOVA only, (ii) SHAP only, and (iii) the proposed ANOVA + SHAP fusion

Variant	Accuracy	Recall	F1	LIME time (s)
XGBoost (All features)	0.9998	0.9995	0.9997	–
[2pt] XGBoost + ANOVA Top-10	0.9839	0.9778	0.9838	2.40
XGBoost + SHAP Top-10	0.9999	0.9997	0.9999	2.57
XGBoost + ANOVA + SHAP Top-10	0.9998	0.9995	0.9997	2.44

Table 20 Ablation study on PFCP Session Deletion traffic. Top-10 variants are obtained with (i) ANOVA only, (ii) SHAP only, and (iii) the proposed ANOVA + SHAP fusion

Variant	Accuracy	Recall	F1	LIME time (s)
XGBoost (All features)	0.8760	0.7492	0.8566	–
[2pt] XGBoost + ANOVA Top-10	0.8631	0.7262	0.8414	1.55
XGBoost + SHAP Top-10	0.8760	0.7492	0.8566	1.46
XGBoost + ANOVA + SHAP Top-10	0.8760	0.7492	0.8566	1.45

Table 21 Ablation study on PFCP Session Establishment traffic. Top-10 variants are obtained with (i) ANOVA only, (ii) SHAP only, and (iii) the proposed ANOVA + SHAP fusion

Variant	Accuracy	Recall	F1	LIME time (s)
XGBoost (All features)	0.9911	1.0000	0.9911	–
[2pt] XGBoost + ANOVA Top-10	0.9952	1.0000	0.9952	1.83
XGBoost + SHAP Top-10	0.9911	1.0000	0.9911	1.76
XGBoost + ANOVA + SHAP Top-10	0.9911	1.0000	0.9911	1.80

= 0.9969). Here, the fusion variant provides slightly faster explanations (2.16 s vs. 2.30–2.33 s), highlighting that combining statistical filtering with model-driven importance can yield efficiency gains without impairing quality.

Slowloris (Table 19) presents the most challenging case, as ANOVA-only reduces performance more substantially ($F1 = 0.9838$). By contrast, SHAP-only and ANOVA+SHAP both sustain nearly perfect detection ($F1 = 0.9999$ and 0.9997 , respectively). The fusion variant shortens LIME explanation time (2.44 s) compared to SHAP-only (2.57 s), again demonstrating that combining ANOVA with SHAP helps remove redundant features, stabilizing interpretability efficiency while preserving accuracy.

Turning to 5 G traffic, the Session Deletion attack (Table 20) proves especially difficult to classify, as the overall baseline accuracy is lower ($F1 = 0.8566$). Here, SHAP-only and ANOVA+SHAP fusion replicate the full-feature performance, while ANOVA-only reduces $F1$ to 0.8414. Efficiency gains are also observed, as explanation times drop from 1.76 s (baseline equivalent) to 1.45–1.55 s. In contrast, Session Establishment (Table 21) achieves consistently strong performance, with ANOVA-only slightly boosting the $F1$ Score (0.9952) compared to the baseline (0.9911), while SHAP-only and ANOVA+SHAP fusion maintain parity. Explanation times remain efficient (1.76–1.83 s), confirming that the fusion approach does not introduce overhead and is effective even under 5 G control-plane traffic.

In summary, across all six attack types, ANOVA-only consistently risks measurable accuracy losses, particularly in more complex traffic such as Slowloris and PFCP Session Deletion. By contrast, SHAP-only and the proposed ANOVA+SHAP fusion consistently match or exceed the baseline detection quality, with the fusion variant providing modest but consistent gains in explanation efficiency. These findings confirm that combining statistical filtering with model-driven importance yields the best balance between accuracy and interpretability efficiency across diverse networking environments.

5 Conclusion

This paper introduces an efficient and explainable XAI-based IDS framework that mitigates the “black-box” limitations of conventional AI-IDS by integrating statistical feature selection (ANOVA) with SHAP and LIME, thereby enhancing interpretability while preserving high detection performance and computational efficiency. The two-stage method not only reduces the feature space by $\approx 70\%$ but also enforces consistency between SHAP and LIME local explanations, validating the reliability of the interpretations. A comprehensive evaluation on the CIC-DDoS2019, CICIOT2023, and 5 G

PFCP datasets demonstrates robust attack-detection capability across diverse networking environments and attack scenarios: the framework maintains an $F1$ Score $\geq 99\%$ and accelerates LIME explanation generation by up to 87 % with the reduced feature set. Although a full head-to-head benchmark with the latest deep learning IDSs (e.g., DeepIDS and SAIDD) is left for future work, their published detection performance results are already matched or exceeded by our lightweight XGBoost pipeline.

In future work, we will strengthen resilience to noisy data by integrating noise-robust preprocessing (e.g., Isolation Forest for outlier removal, Hampel and median filtering for bursty artefacts, and RobustScaler to temper heavy-tailed feature distributions) and by injecting controlled Gaussian and burst noise during training for data-level augmentation. To counter adversarial manipulation, we will extend the training loop with Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD) adversarial examples, followed by adversarial re-training and a lightweight detector based on Mahalanobis distance to flag perturbed flows at run-time. We also plan to develop a real-time online version and explore federated deployment for privacy-preserving scalability, while adding a packet-level module that leverages 1-D CNN embeddings so that raw packet traces can be processed alongside flow features.

Author contributions BN organized the paper’s structure, formulated the state-of-the-art, defined the goals, and designed the experiments. BN and AVJ collaborated on formulating the framework, with TB reviewing and finalizing it. AVJ conducted the initial experiments using the CIC-DDoS2019 dataset, which were later expanded by BN to include the CICIOT 2023 and 5 G PFCP datasets. All authors reviewed the manuscript.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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